Useful Memory under Drift

Temporal-Validity Horizons for Finite-Memory Distribution Tracking

Vinicius Parreira

Independent Researcher

`figueiredo@pm.me`

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Abstract

We study how long retained evidence remains statistically valid for distribution tracking under drift. Retaining more data reduces estimation noise but increases temporal mismatch, so finite memory has a nontrivial validity horizon. This horizon is the scale at which the finite-sample term and the drift-induced staleness term balance, yielding both an optimal memory law and a useful-memory region whose normalized shape depends only on the finite-sample and drift exponents. The law closes in a one-dimensional root-n proof anchor and in theorem-level embedded Sinkhorn classes, including bounded-cardinality finite-discrete models and a calibrated support-growth family; finite projection families and sliced transport inherit the same structure under uniform projected control. Lag-geometry inference then yields identifiability and quadratic regret guarantees for data-driven memory selection. Experiments exhibit the predicted U-curves, roughness-dependent scaling, real-stream finite-memory effects, and validity loss before detector alarm.

Keywords: tracking under drift; temporal validity; finite-memory tracking; distribution tracking; Wasserstein distance; Sinkhorn divergence.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

A drifting system is governed by two coupled clocks. The tracking clock rewards retention because larger windows reduce sampling noise. The validity clock penalizes retention because older evidence ceases to represent the present distribution. Change detection sharpens the same tension: a detector can remain silent even after retained evidence has already become stale. The central question is therefore not only when change becomes detectable, but how long retained evidence remains valid for distribution tracking.

Let P_t denote the present target law, let $\bar{P}_t^{(n)}$ denote the window target formed from the last n target laws, and let $\widehat{P}_t^{(n)}$ be the estimator built from the last n observations. Tracking error then separates into a finite-sample term relative to $\bar{P}_t^{(n)}$ and a staleness term between $\bar{P}_t^{(n)}$ and P_t:

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^{H}.$$

When the finite-sample term decays like n^{-a} and the staleness term grows like ζn^H, their balance defines the temporal-validity horizon and the induced useful-memory region.

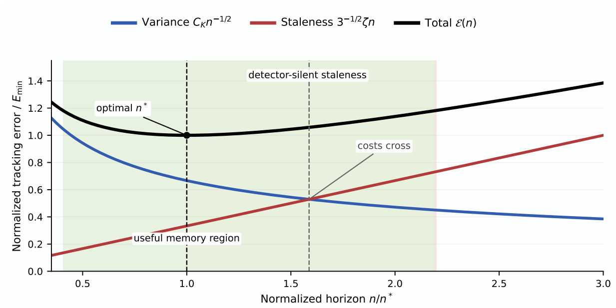

Figure 1: Temporal-validity horizon under drift. The finite-sample term falls with horizon, the staleness term rises with age, and total error is minimized at the temporal-validity horizon n^*. The green band marks a near-optimal useful-memory region, and the red band marks pre-detection validity loss: retained evidence can stop representing the present before enough changepoint evidence accumulates to trigger an alarm.

The finite-sample exponent a depends on the geometry used to compare distributions, while the roughness exponent H and amplitude ζ describe how quickly retained evidence loses temporal validity. The resulting horizon is structural: it is imposed jointly by metric geometry and drift regularity. This is why dimensionality enters the problem only through the finite-sample regime supplied by the chosen geometry: a poor exponent collapses the useful memory scale, while projection, smoothing, or regularization can preserve it.

Three theorem-level realizations make the law concrete. A one-dimensional root-n proof anchor closes the basic finite-memory regime, while bounded- cardinality finite-discrete and calibrated support-growth embedded Sinkhorn classes show that the same mechanism persists in operational transport geometry. Gaussian lower and benchmark results identify the horizon as a property of drift geometry itself.

Projection and sliced geometries preserve the same mechanism under transfer, useful-memory localization from lag geometry inherits identifiability and regret bounds, and the same object appears empirically through interior U-curves, roughness scaling, real-stream diagnostics, and validity loss before detector alarm.

2 Temporal-Validity Horizon

2.1 General Horizon Law

Theorem 2.1 (Abstract upper law)**.** *Let d be a probability metric satisfying the triangle inequality. At time t, let P_t be the present target law, let $\bar{P}_t^{(n)}$ be the window target, and let $\widehat{P}_t^{(n)}$ be the estimator built from the last n observations. Assume that for some constants $C_K > 0$, $C_S > 0$, exponent $a > 0$, roughness index $H \in (0, 1]$, and amplitude $\zeta > 0$,*

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \leq C_K n^{-a}$$

and

$$d\Big(\bar{P}_t^{(n)}, P_t\Big) \leq C_S \zeta n^H$$

hold for all n in the regime of interest. Then

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \leq C_K n^{-a} + C_S \zeta n^H.$$

Proof. By the triangle inequality,

$$d\Big(\widehat{P}_t^{(n)}, P_t\Big) \leq d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) + d\Big(\bar{P}_t^{(n)}, P_t\Big).$$

Taking expectations and applying the two assumptions yields the claim.

The path class supplies the staleness term, while the chosen geometry and sample regime supply the finite-sample term.

Proposition 2.2 (Triangular-array transfer for Hadamard-differentiable functionals)**.** *Let $\mathcal{T}$ be a functional on probability laws that is Hadamard differentiable at P_t tangentially to a linear subspace of signed measures, with derivative $\dot{\mathcal{T}}_{P_t}$ bounded on the relevant tangent class. Suppose that for the triangular-array estimator and its window target, the base metric satisfies*

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \leq C_K n^{-a}$$

and the drift envelope satisfies

$$d\Big(\bar{P}_t^{(n)}, P_t\Big) \leq C_S \zeta n^H.$$

If the remainder in the Hadamard expansion is $o(n^{-a})$ uniformly on the relevant class, then there exist constants $\widetilde{C}_K, \widetilde{C}_S > 0$ such that

$$\mathbb{E}\, \big\| \mathcal{T}(\widehat{P}_t^{(n)}) - \mathcal{T}(P_t) \big\| \leq \widetilde{C}_K n^{-a} + \widetilde{C}_S \zeta n^H.$$

Proof. Write the Hadamard expansion at P_t and apply the boundedness of the derivative to the centered triangular-array fluctuation. The drift term contributes through the base-metric envelope because $\mathcal{T}$ is locally Lipschitz on the class. The remainder is lower order by assumption, so the transformed law inherits the horizon envelope.

Proposition 2.3 (Finite-dimensional smooth transfer)**.** *Let $\phi_1, \ldots, \phi_k$ be bounded measurable functions and let $Z(P) = \big(\mathbb{E}_P[\phi_1], \ldots, \mathbb{E}_P[\phi_k]\big)$. Let $g \in C^2(U)$ on an open set $U \subset \mathbb{R}^k$ containing the relevant range of Z. Define $T(P) = g(Z(P))$. Suppose that on the triangular-array window target and estimator,*

$$\mathbb{E}\, \| Z(\widehat{P}_t^{(n)}) - Z(\bar{P}_t^{(n)}) \| \leq C_K n^{-a}$$

and

$$\| Z(\bar{P}_t^{(n)}) - Z(P_t) \| \leq C_S \zeta n^H.$$

If the Hessian of g is bounded by M on the relevant range, then

$$\mathbb{E}\, | T(\widehat{P}_t^{(n)}) - T(P_t) | \leq \widetilde{C}_K n^{-a} + \widetilde{C}_S \zeta n^H,$$

and the Taylor remainder is quadratic in the moment error:

$$\begin{aligned} &|T(P') - T(P) - \nabla g(Z(P))^\top \big(Z(P') - Z(P)\big)| \\ &\quad \leq \frac{M}{2} \| Z(P') - Z(P) \|^2. \end{aligned}$$

Proof. Apply the multivariate Taylor formula to g around $Z(P_t)$. The bounded Hessian gives the stated quadratic remainder. The first-order term is controlled by the assumed finite-sample and drift envelopes for Z, and the argument from proposition 2.2 yields the horizon law for T.

Corollary 2.4 (Temporal-validity horizon)**.** *Under the assumptions of theorem 2.1, the upper envelope*

$$\Phi(n) := C_K n^{-a} + C_S \zeta n^H$$

is minimized, up to constant factors depending only on a, H, and C_S, at

$$n^*(a, H) \asymp (C_K/\zeta)^{1/(a+H)},$$

and the corresponding error scale is

$$\Phi(n^*) \asymp C_K^{H/(a+H)} \zeta^{a/(a+H)}.$$

Proof. Balance the decreasing finite-sample term against the increasing staleness term:

$$C_K n^{-a} \asymp C_S \zeta n^H.$$

This gives $n^{a+H} \asymp C_K/\zeta$, hence the displayed horizon law. Substituting that scale back into either term gives the optimal error scale.

Write the continuous optimizer of Φ as

$$n_\star := \left(\frac{a C_K}{H C_S \zeta} \right)^{1/(a+H)}.$$

This optimizer differs from the scale $n^*(a, H)$ only by constants depending on a, H, and C_S, so the asymptotic horizon law remains unchanged.

2.2 Useful-Memory Geometry

The horizon scale induces a near-optimal region of memory lengths. For any $\delta > 0$, define the useful-memory region by

$$\mathcal{U}_\delta := \{ n > 0 : \Phi(n) \leq (1 + \delta) \Phi(n_\star) \}.$$

This object records how much memory remains acceptable up to a fixed tolerance.

Proposition 2.5 (Useful-memory region at tolerance δ)**.** *Let $\Phi(n) = C_K n^{-a} + C_S \zeta n^H$ with $a, H > 0$, and let $n_\star$ be its continuous optimizer. Then the normalized profile*

$$\Psi(x) := \frac{\Phi(n_\star x)}{\Phi(n_\star)} = \frac{H x^{-a} + a x^H}{a + H}$$

is strictly decreasing on $(0,1]$ *and strictly increasing on* $[1,\infty)$*. Consequently,*

$$\mathcal{U}_\delta = [n_-, n_+] = n_\star[x_-(\delta;a,H), x_+(\delta;a,H)],$$

where $x_-(\delta;a,H) \le 1 \le x_+(\delta;a,H)$ *are the two solutions of* $\Psi(x) = 1+\delta$*. In particular, the useful-memory region scales linearly with the horizon scale* $n_\star$*, and its relative shape depends only on* a*,* H*, and* δ*.*

Proof. Since $n_\star$ minimizes Φ, differentiation gives

$$aC_K n_\star^{-a} = HC_S\zeta n_\star^H.$$

Substituting $n = n_\star x$ into Φ and dividing by $\Phi(n_\star)$ yields the displayed identity for Ψ. Differentiation then gives

$$\Psi'(x) = \frac{aH}{a+H}\left(x^{H-1} - x^{-a-1}\right),$$

which is negative for $x<1$ and positive for $x>1$. Since $\Psi(1)=1$ and $\Psi(x)\to\infty$ as $x\to 0$ or $x\to\infty$, the sublevel set $\{x>0 : \Psi(x)\le 1+\delta\}$ is a compact interval containing 1, which proves the representation of $\mathcal{U}_\delta$.

For small tolerance, the useful-memory region has the local expansion

$$x_\pm(\delta;a,H) = 1 \pm \sqrt{\frac{2\delta}{aH}} + o(\sqrt{\delta}), \qquad \delta\downarrow 0,$$

because $\Psi''(1) = aH$. The local curvature therefore controls how quickly the near-optimal region contracts around the horizon.

The horizon depends jointly on the finite-sample rate exponent and the roughness exponent. Different geometries and different drift regularities induce different memory scales.

2.3 Drift Regularity and Staleness Control

The staleness side of the law is controlled by a deterministic path class. For $H\in(0,1]$ and $\zeta>0$, write

$$\mathfrak{P}_H(\zeta) := \left\{(P_s)_{s\in\mathbb{Z}} : W_2(P_{t-j}, P_t) \le \zeta j^H \text{ for all } j\ge 0\right\}.$$

The case $H=1$ is the local Lipschitz envelope; smaller H corresponds to slower accumulation of staleness with lag.

Lemma 2.6 (Hölder staleness bound via averaged couplings)**.** *Let*

$$\bar P_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} P_{t-j}.$$

If the target path belongs to $\mathfrak{P}_H(\zeta)$*, then*

$$W_2^2\left(\bar P_t^{(n)}, P_t\right) \le \frac{\zeta^2}{n}\sum_{j=0}^{n-1} j^{2H},$$

and therefore, by Jensen's inequality,

$$W_2\left(\bar P_t^{(n)}, P_t\right) \le \zeta\left(\frac{1}{n}\sum_{j=0}^{n-1} j^{2H}\right)^{1/2} = C_{H,n}\,\zeta n^H,$$

where

$$C_{H,n} := \left(n^{-(2H+1)}\sum_{j=0}^{n-1} j^{2H}\right)^{1/2}.$$

In particular, $C_{H,n}\to(2H+1)^{-1/2}$ *as* $n\to\infty$*.*

Proof. For each lag j, let π_j be an optimal coupling between P_{t-j} and P_t. Their average

$$\bar\pi := \frac{1}{n}\sum_{j=0}^{n-1}\pi_j$$

couples $\bar P_t^{(n)}$ with P_t, so

$$\begin{aligned} W_2^2\left(\bar P_t^{(n)}, P_t\right) &\le \int \|x-y\|^2\, d\bar\pi(x,y) \\ &= \frac{1}{n}\sum_{j=0}^{n-1} W_2^2(P_{t-j}, P_t)\,. \end{aligned}$$

Applying the Hölder envelope gives the displayed finite-n constant, and taking square roots completes the proof.

Combining lemma 2.6 with corollary 2.4 yields the horizon family

$$n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)}.$$

In the root-n finite-sample regime $a=1/2$,

$$n^*(1/2,H) \asymp (C_K/\zeta)^{2/(1+2H)}.$$

The finite-n constant in lemma 2.6 is the constant used here for uniform windows; the simpler $(2H+1)^{-1/2}$ expression is its large-window limit.

2.4 One-Dimensional Proof Anchor

The bounded-support fixed-span one-dimensional triangular array provides a tractable rigorous finite-sample instantiation of the abstract law. The model is built on common compact support, uniformly regular window mixtures, and an integrated Bahadur remainder hypothesis.

Proposition 2.7 (Root-n finite-sample bound in the 1D proof model)**.** *Let* $X_{n,1},\dots,X_{n,n}$ *be independent one-dimensional observations with laws* $P_{n,1},\dots,P_{n,n}$*, and define the window mixture*

$$\bar P_n = \frac{1}{n}\sum_{j=1}^n P_{n,j}.$$

Assume:

(i) all $P_{n,j}$ *are supported on a common compact interval* $[L,U]$*;*

(ii) the within-window drift span is uniformly bounded in n*;*

(iii) each mixture law $\bar P_n$ *has a density* $\bar f_n$ *on* $[L,U]$ *that is bounded below by a positive constant and uniformly Hölder;*

(iv) writing q_n for the quantile function of $\bar{P}_n$ and $\widehat{q}_n$ for the empirical quantile function of $\widehat{P}_n^{\mathrm{tri}}$, one has the Bahadur representation

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u), \qquad u \in (0,1),$$

with

$$\mathbb{E}\int_0^1 R_n(u)^2\,du = o(n^{-1}).$$

Then

$$\mathbb{E}\,W_2^2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = O(n^{-1}),$$

and therefore, by Jensen's inequality,

$$\mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = O(n^{-1/2}).$$

Proof. In one dimension,

$$W_2^2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = \int_0^1 \big(\widehat{q}_n(u) - q_n(u)\big)^2\,du,$$

so the problem reduces to the integrated L_2 error of the empirical quantile process. By item (iv),

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u).$$

Define

$$Z_n(u) := \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))}.$$

Then

$$\mathbb{E}\int_0^1 \big(\widehat{q}_n(u) - q_n(u)\big)^2 du = \mathbb{E}\int_0^1 \big(Z_n(u) + R_n(u)\big)^2 du.$$

Let $m > 0$ be a uniform lower bound for $\bar{f}_n$. Since

$$\widehat{F}_n(x) = \frac{1}{n}\sum_{j=1}^n \mathbf{1}\{X_{n,j} \le x\},$$

independence across the row gives

$$\mathrm{Var}(\widehat{F}_n(q_n(u))) = \frac{1}{n^2}\sum_{j=1}^n p_{n,j}(u)(1 - p_{n,j}(u)) \le \frac{1}{4n},$$

where $p_{n,j}(u) = P_{n,j}((-\infty, q_n(u)])$. Therefore

$$\mathbb{E}\int_0^1 Z_n(u)^2 du \le m^{-2}\int_0^1 \mathrm{Var}(\widehat{F}_n(q_n(u)))du \le \frac{1}{4m^2 n}.$$

By item (iv),

$$\mathbb{E}\int_0^1 R_n(u)^2 du = o(n^{-1}).$$

The cross term is then $o(n^{-1})$ by Cauchy–Schwarz, since $\mathbb{E}\int_0^1 Z_n(u)^2 du = O(n^{-1})$ and $\mathbb{E}\int_0^1 R_n(u)^2 du = o(n^{-1})$. Hence the expected squared Wasserstein error is $O(n^{-1})$, and Jensen's inequality gives the root-n rate for $\mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big)$.

The main conclusion is the exponent. Under this bounded-support fixed-span regime, the triangular-array finite-sample term remains root-n.

2.5 Verifiable Smooth Deformation Class

An explicit deformation class makes the proof-model hypotheses verifiable. Let

$$g_\delta(x) = x + \delta x(1-x), \qquad x \in [0,1], \qquad |\delta| \le \Delta < 1.$$

Then g_δ is an increasing C^2 diffeomorphism of $[0,1]$, with

$$1 - \Delta \le g_\delta'(x) \le 1 + \Delta, \qquad |g_\delta''(x)| \le 2\Delta,$$

so the transformed densities satisfy

$$\frac{1}{1+\Delta} \le f_\delta(y) \le \frac{1}{1-\Delta},$$

and the same support is preserved.

Corollary 2.8 (Verifiable Bahadur remainder for the smooth deformation class)**.** *Assume that each row law has the form*

$$P_{n,j} = (g_{\delta_{n,j}})_\# \,\mathrm{Unif}[0,1], \qquad |\delta_{n,j}| \le \Delta < 1,$$

and that the within-window span is uniformly bounded in n. Then item (iv) of proposition 2.7 holds. More precisely, there exists a constant $C_\Delta < \infty$ such that

$$\mathbb{E}\int_0^1 R_n(u)^2\,du \le C_\Delta n^{-2}.$$

Consequently,

$$\mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = O(n^{-1/2})$$

on this deformation class.

Proof. Write $\bar{F}_n$ and q_n for the mixture cdf and quantile. Since each $g_{\delta_{n,j}}$ is an increasing diffeomorphism of $[0,1]$, every mixture law $\bar{P}_n$ is supported on $[0,1]$ and has density $\bar{f}_n = \frac{1}{n}\sum_{j=1}^n f_{\delta_{n,j}}$ satisfying the explicit bounds

$$\frac{1}{1+\Delta} \le \bar{f}_n(y) \le \frac{1}{1-\Delta}, \qquad \|\bar{f}_n'\|_\infty \le \frac{2\Delta}{(1-\Delta)^3}.$$

Hence the inverse map $\bar{F}_n^{-1}$ is uniformly twice differentiable on $(0,1)$, with second derivative bounded by a constant depending only on Δ. Applying the second-order Taylor expansion of $\bar{F}_n^{-1}$ around $u = \bar{F}_n(q_n(u))$ gives

$$\widehat{q}_n(u) - q_n(u) = \frac{u - \widehat{F}_n(q_n(u))}{\bar{f}_n(q_n(u))} + R_n(u),$$

with

$$|R_n(u)| \le C_\Delta\,|\widehat{F}_n(q_n(u)) - u|^2.$$

Now

$$\widehat{F}_n(q_n(u)) = \frac{1}{n}\sum_{j=1}^n \mathbf{1}\{X_{n,j} \le q_n(u)\}$$

is an average of independent Bernoulli variables with mean u. Uniformly in u, its fourth centered moment is $O(n^{-2})$. Therefore

$$\mathbb{E}\,R_n(u)^2 \le C_\Delta\,\mathbb{E}|\widehat{F}_n(q_n(u)) - u|^4 \le C_\Delta' n^{-2},$$

and integrating over $u \in (0,1)$ yields

$$\mathbb{E}\int_0^1 R_n(u)^2\,du \le C_\Delta' n^{-2} = o(n^{-1}).$$

The root-n claim then follows from proposition 2.7. The same smoothness package is consistent with the classical Bahadur theory of Bahadur [1966] and van der Vaart [1998, Chapter 21], but here the explicit deformation bounds make the integrated remainder quantitative.

Remark 2.9 (Weak rowwise dependence). The proof of proposition 2.7 uses independence within a row only through the $O(n^{-1})$ variance bound for the empirical cdf term and through item (iv). Consequently, the same root-n conclusion persists under any short-range dependent rowwise model for which the empirical cdf still has L_2 fluctuation of order $n^{-1/2}$ and the integrated Bahadur remainder remains $o(n^{-1})$. Short-range mixing conditions are one natural route to such a package, but no general dependent-row theorem.

Empirical diagnostics for the smooth deformation class and the fixed-support triangular-array model appear in appendix B.

2.6 Structural Lower Bound

The upper law would be substantially weaker if it were only an estimator-specific statement. The lower bound shows that, in the root-n finite-sample regime $a = 1/2$, the horizon is imposed by the drift geometry itself.

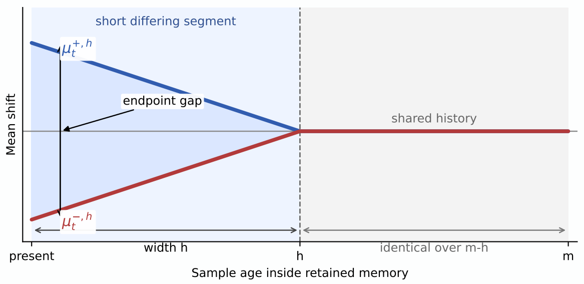

Figure 2: Least-favorable Gaussian path pair for the root-n horizon lower bound. The two paths share most of the retained history and differ only on a recent segment of width h, yet remain separated at the present. That recent segment already produces the same horizon exponent as the upper law in the root-n regime.

Proposition 2.10 (Structural lower bound in the root-n regime)**.** *Fix $H \in (0, 1]$ and work on the Gaussian location subclass with known variance σ^2 and path pair*

$$\mu_{t-j}^{\pm,h} = \pm\beta(h-j)_+^H, \qquad j = 0, 1, \ldots, h,$$

where $\beta \le \zeta$ enforces the roughness budget. Let $\mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}$ denote the corresponding Gaussian location subclass of target paths, and suppose the retained observations are independent with laws $N(\mu_{t-j}^{\pm,h}, \sigma^2)$. Then there exists a constant $c_H > 0$ depending only on H such that the endpoint tracking risk over this subclass satisfies

$$\inf_{\widehat{P}} \sup_{P \in \mathcal{G}_{H,\zeta}^{\mathrm{Gauss}}} \mathbb{E}\, W_2\left(\widehat{P}_t, P_t\right) \ge c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Consequently, in the root-n finite-sample regime $a = 1/2$, the horizon scale

$$h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$$

is structural even on this restricted subclass.

Proof. Compare two hypotheses generated by the path pair $\mu_{t-j}^{+,h}$ and $\mu_{t-j}^{-,h}$. At the present time, the endpoint separation is of order βh^H. On the Gaussian location subclass, this is also the W_2 separation. The Kullback–Leibler divergence between the two hypotheses is proportional to

$$\beta^2 \sum_{r=1}^{h} r^{2H}/\sigma^2.$$

Choosing

$$\beta = \min\left\{\zeta, \frac{\sigma}{2\sqrt{\sum_{r=1}^{h} r^{2H}}}\right\}$$

keeps the testing problem in the Le Cam regime while respecting the roughness budget. The resulting two-point lower bound is of order βh^H. Balancing the active roughness constraint against the testing constraint gives $h^*(H) \asymp (\sigma/\zeta)^{2/(2H+1)}$ and therefore the displayed lower law. Appendix appendix A gives the exact Gaussian two-point calculation used here.

The lower bound is subclass based and not class-tight, but it is enough to show that the horizon is a structural feature of the problem and not an estimator artifact.

The proposition above is the lower-bound fact needed for the main horizon law. The same Gaussian construction also exposes the constant-level geometry. The following proposition gives the sharp Gaussian constants available within that proof scheme.

Appendix appendix A records the sharpened Gaussian constant geometry for the ramp and endpoint-minimal profiles.

The same lower-bound geometry also yields a full class-level benchmark once the model is specialized to Gaussian location tracking. In that setting, the deterministic Hölder envelope, the finite-memory estimator, and the lower-bound construction all live in the same one-dimensional parameterization, so this model admits a direct minimax rate statement.

Theorem 2.11 (Gaussian location minimax benchmark on the deterministic Hölder class)**.** *Fix $H \in (0, 1]$, $\zeta > 0$, and $\sigma > 0$. Let $\mathfrak{M}_H(\zeta)$ denote the class of deterministic mean paths $(\mu_s)_{s\in\mathbb{Z}}$ such that*

$$|\mu_{t-j} - \mu_t| \le \zeta j^H \qquad \text{for all } j \ge 0.$$

Suppose the observations are independent with

$$Y_{t-j} \sim N(\mu_{t-j}, \sigma^2), \qquad j = 0, 1, \ldots, n-1,$$

and let the target law at time t be $P_t = N(\mu_t, \sigma^2)$. Define the optimal finite-memory minimax risk

$$\mathcal{R}_H^*(\sigma, \zeta) := \inf_{n\ge1} \inf_{\widehat{P}_t^{(n)}} \sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\left(\widehat{P}_t^{(n)}, P_t\right).$$

Then there exist constants $0 < c_H \le C_H < \infty$, depending only on H, such that

$$c_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)} \le \mathcal{R}_H^*(\sigma, \zeta) \le C_H\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

In particular, the deterministic Hölder class has horizon scale

$$n_H^* \asymp (\sigma/\zeta)^{2/(2H+1)}.$$

Moreover, the uniform-window mean estimator has an exact asymptotic upper constant on this benchmark. If $y_H > 0$ solves

$$Hy_H\big(2\Phi_{\mathrm{G}}(y_H) - 1\big) = \varphi_{\mathrm{G}}(y_H),$$

then its asymptotic horizon shape is

$$A_H^{\mathrm{mean}} = ((H+1)y_H)^{2/(2H+1)}$$

and its asymptotic risk constant is

$$U_H^{\mathrm{mean}} = ((H+1)y_H)^{-1/(2H+1)} \times \Big(2\varphi_{\mathrm{G}}(y_H) + y_H\big(2\Phi_{\mathrm{G}}(y_H) - 1\big)\Big).$$

Proof. For the upper bound, estimate the present mean by the uniform-window average

$$\widehat{\mu}_t^{(n)} := \frac{1}{n}\sum_{j=0}^{n-1} Y_{t-j}$$

and set $\widehat{P}_t^{(n)} = N(\widehat{\mu}_t^{(n)}, \sigma^2)$. Because equal-variance Gaussian laws satisfy

$$W_2\big(N(m_1,\sigma^2), N(m_2,\sigma^2)\big) = |m_1 - m_2|,$$

the risk reduces to mean estimation error. Since

$$\widehat{\mu}_t^{(n)} - \frac{1}{n}\sum_{j=0}^{n-1}\mu_{t-j} \sim N\Big(0, \frac{\sigma^2}{n}\Big),$$

the noise term is $\mathbb{E}|N(0,\sigma^2/n)| = \sigma\sqrt{2/(\pi n)}$, while the deterministic bias is bounded by

$$\frac{\zeta}{n}\sum_{j=0}^{n-1} j^H,$$

so

$$\sup_{\mu\in\mathfrak{M}_H(\zeta)} \mathbb{E}\, W_2\Big(\widehat{P}_t^{(n)}, P_t\Big) \le \sigma\sqrt{\frac{2}{\pi n}} + \frac{\zeta}{n}\sum_{j=0}^{n-1} j^H.$$

Rescaling by $n = A(\sigma/\zeta)^{2/(2H+1)}$ yields a one-parameter normalized upper envelope. Differentiating its exact Gaussian-location form gives the stated root equation for y_H, and therefore the displayed asymptotic upper constant and horizon shape. In particular, this proves the stated upper rate and horizon scale.

For the lower bound, use the symmetric two-point subfamily

$$\mu_{t-j}^{\pm,h} = \pm\beta\big(h^H - j^H\big)_+, \qquad j \ge 0,$$

with $\beta \le \zeta$. This family lies in $\mathfrak{M}_H(\zeta)$ because

$$|\mu_t^{\pm,h} - \mu_{t-j}^{\pm,h}| = \beta\min(h^H, j^H) \le \zeta j^H.$$

Moreover, for fixed endpoint value βh^H, it minimizes the Gaussian testing energy over all symmetric lag profiles allowed by the present-point Hölder envelope, since each lag coordinate is independently minimized by $\beta(h^H - j^H)_+$. The exact Gaussian two-point calculation from the lower-bound analysis in appendix A therefore applies on the full deterministic Hölder class and yields a lower bound of order

$$\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}.$$

Minimizing over h gives the same horizon scale $(\sigma/\zeta)^{2/(2H+1)}$. Combining the two sides proves the theorem.

2.7 Projection and Sliced Transfer

The horizon law also extends beyond one dimension when the multivariate geometry reduces to a uniformly controlled family of one-dimensional transport problems. This is the natural setting for sliced or coordinate-averaged Wasserstein constructions, where the finite-sample term is built by integrating projected one-dimensional errors.

Corollary 2.12 (Finite-family projection transfer)**.** *Let $\Pi = \{\theta_1, \dots, \theta_m\}$ be a finite family of linear maps from the ambient space to $\mathbb{R}$, and define*

$$d_\Pi(P,Q) := \Big(\frac{1}{m}\sum_{r=1}^{m} W_2^2(P_{\theta_r}, Q_{\theta_r})\Big)^{1/2}.$$

Assume that for every $r = 1, \dots, m$:

(i) the projected triangular-array window satisfies the hypotheses of proposition 2.7 with constants uniform in r;

(ii) the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in r.

Then there exist constants $C_{K,\Pi}, C_{S,\Pi} < \infty$ such that

$$\mathbb{E}\, d_\Pi\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_{K,\Pi} n^{-1/2} + C_{S,\Pi}\zeta n^H.$$

Consequently,

$$n_\Pi^*(H) \asymp (C_{K,\Pi}/\zeta)^{1/(H+1/2)}.$$

In particular, the same law holds for coordinate-averaged product constructions obtained by taking Π to be the coordinate projections.

Proof. For each r, proposition 2.7 gives

$$\mathbb{E}\, W_2^2\Big(\widehat{P}_{t,\theta_r}^{(n)}, \bar{P}_{t,\theta_r}^{(n)}\Big) \le Cn^{-1}$$

with C uniform in r. Averaging over r yields

$$\mathbb{E}\, d_\Pi^2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) \le Cn^{-1},$$

and Jensen gives the root-n bound for $\mathbb{E}\, d_\Pi(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound averages in the same way:

$$d_\Pi\Big(\bar{P}_t^{(n)}, P_t\Big) \le C_{S,\Pi}\zeta n^H.$$

Applying theorem 2.1 with $d = d_\Pi$ yields the displayed envelope and horizon scale.

Corollary 2.13 (Sliced-Wasserstein transfer under uniform projected control)**.** *Let ρ be a probability measure on a family of one-dimensional projections θ, and define*

$$SW_{2,\rho}^2(P,Q) := \int W_2^2(P_\theta, Q_\theta)\,\rho(d\theta),$$

where P_θ denotes the pushforward of P under projection θ. Assume that for every θ in the support of ρ:

(i) the projected triangular-array window satisfies the hypotheses of proposition 2.7 with constants uniform in θ;

(ii) the projected drift path obeys the same Hölder staleness envelope with exponent H and amplitude at most ζ uniformly in θ.

Then there exist constants $C_K, C_S < \infty$ such that

$$\mathbb{E}\, SW_{2,\rho}\left(\widehat{P}_t^{(n)}, P_t\right) \le C_K n^{-1/2} + C_S \zeta n^H.$$

Consequently,

$$n^*(H) \asymp (C_K/\zeta)^{1/(H+1/2)}.$$

Finite projection families are included by taking ρ supported on finitely many atoms; corollary 2.12 states that case directly for multivariate tracking.

Proof. For each projection θ, proposition 2.7 gives

$$\mathbb{E}\, W_2^2\left(\widehat{P}_{t,\theta}^{(n)}, \bar{P}_{t,\theta}^{(n)}\right) \le Cn^{-1}$$

with C uniform in θ. Integrating over ρ yields

$$\mathbb{E}\, SW_{2,\rho}^2\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) \le Cn^{-1},$$

and Jensen gives the root-n bound for $\mathbb{E}\, SW_{2,\rho}(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)})$. The projected Hölder bound integrates in the same way, giving

$$SW_{2,\rho}\left(\bar{P}_t^{(n)}, P_t\right) \le C_S \zeta n^H.$$

Applying theorem 2.1 with $d = SW_{2,\rho}$ yields the stated envelope and horizon scale.

Weighted-memory summaries and the effective-sample-size comparison are recorded in appendix A.

2.8 Embedded Sinkhorn Theorems

The Gaussian benchmark closes a full path-class model, while the fixed-ε Sinkhorn results place the same horizon logic in an operational transport geometry. The linear-Gaussian entropic-OT setting suggests an intermediate recursive regime between the i.i.d. fixed-ε theory and the embedded low-dimensional extension [Kalman, 1960, Jazwinski, 1970, Akyildiz et al., 2024].

Fixed-ε Sinkhorn setting. For fixed-ε debiased Sinkhorn, the central question is whether a practical geometry can supply a stable finite-sample term on the embedded fixed-span model. The target law is

$$\mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_\varepsilon n^{-a_\varepsilon}$$

under bounded support, fixed span, and structural low-dimensionality, where S_ε denotes a fixed-ε Sinkhorn-type geometry and the exponent a_ε matches the corresponding i.i.d. mixture benchmark.

Definition 2.14 (Fixed-ε Sinkhorn setting)**.** Fix a bounded embedded fixed-span model, an intrinsic complexity index k, a dual Hölder smoothness level α, and an operational band $\mathcal{B} \subset (0, \infty)$ of regularization values. We say that this model is in the fixed-ε Sinkhorn setting if:

(i) the i.i.d. fixed-ε benchmark on the same support class admits a finite-sample bound with exponent a_ε on $\mathcal{B}$;

(ii) the triangular-window and i.i.d.-mixture supports have identical operational covering complexity on $\mathcal{B}$;

(iii) the dual class lies in the parametric adaptation region $\alpha > k/2$.

Proposition 2.15 (Structural ingredients for fixed-ε Sinkhorn)**.** *On the bounded embedded fixed-span model with squared-Euclidean cost, the first three items in definition 2.14 are structural: the support-complexity inheritance is exact, and the dual-complexity condition is precisely the LCA threshold $\alpha > k/2$.*

Theorem 2.16 (Finite-Discrete Embedded Sinkhorn Horizon Law)**.** *Fix a compact moderate band $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0, \infty)$, a support-size bound $m_0 \in \mathbb{N}$, a support-diameter bound $D < \infty$, and a mass floor $\lambda \in (0, 1/m_0]$. Suppose that for every window (t, n) there exists a support set*

$$X_{t,n} = \{x_1, \ldots, x_m\}, \qquad m \le m_0,$$

such that

$$\bar{P}_t^{(n)} = \sum_{i=1}^m b_i \delta_{x_i}, \qquad \widehat{P}_{t,\mathrm{tri}}^{(n)} = \sum_{i=1}^m \widehat{b}_i \delta_{x_i},$$

with

$$\max_{1 \le i,j \le m} \|x_i - x_j\|^2 \le D^2, \qquad b_i \ge \lambda \quad \textit{for all } i.$$

Then there exists a finite constant $C_{\mathcal{B},m_0,D,\lambda}$ such that

$$\sup_{\varepsilon \in \mathcal{B}} \mathbb{E}\, S_\varepsilon\left(\widehat{P}_{t,\mathrm{tri}}^{(n)}, \bar{P}_t^{(n)}\right) \le C_{\mathcal{B},m_0,D,\lambda} n^{-1/2}.$$

Consequently the corresponding temporal-validity horizon on S_ε has the root-n horizon scale

$$n_{\mathcal{B}}^*(H) \asymp \left(C_{\mathcal{B},m_0,D,\lambda}/\zeta\right)^{1/(H+1/2)}.$$

Proof. Fix (t, n) and abbreviate the support size by m. For each $\varepsilon \in \mathcal{B}$, let K_ε be the Sinkhorn kernel on $X_{t,n}$ associated with the squared-Euclidean cost. Since the support diameter is at most D,

$$K_{ij}(\varepsilon) \ge \exp(-D^2/\varepsilon_{\min}) =: \alpha$$

for all i, j. Let

$$P_{\varepsilon,b}(i,j) := \frac{K_{ij}(\varepsilon) b_j}{\sum_{\ell=1}^m K_{i\ell}(\varepsilon) b_\ell}$$

be the weighted self-coupling block. Because $\sum_\ell K_{i\ell}(\varepsilon) b_\ell \le 1$ and $b_j \ge \lambda$, one has

$$P_{\varepsilon,b}(i,j) \ge \alpha\lambda.$$

Hence

$$P_{\varepsilon,b} = \delta U_m + (1-\delta) Q_{\varepsilon,b}, \qquad \delta := m\alpha\lambda,$$

where U_m is the uniform kernel on m atoms and $Q_{\varepsilon,b}$ is stochastic. Since $m\lambda \le 1$, one has $0 < \delta \le 1$. The centered self-coupling block therefore contracts the zero-mass signed-measure subspace $V_0 := \{z \in \mathbb{R}^m : z\mathbf{1} = 0\}$ in total variation norm: for $z \in V_0$,

$$zP_{\varepsilon,b} = (1-\delta) z Q_{\varepsilon,b}, \qquad \|zP_{\varepsilon,b}\|_1 \le (1-\delta)\|z\|_1.$$

Hence

$$\|zP_{\varepsilon,b}^2\|_1 \le (1-\delta)^2\|z\|_1 \le (1-\alpha\lambda)^2\|z\|_1,$$

uniformly over all admissible windows. Consequently

$$\big\|(I - P_{\varepsilon,b}^2|_{V_0})^{-1}\big\| \le \big[1-(1-\alpha\lambda)^2\big]^{-1}.$$

Because the support size is uniformly bounded and the target weights stay in the compact interior set $\{b \in \Delta_m : b_i \ge \lambda\}$, and because only the finitely many dimensions $m \le m_0$ occur, the finite-dimensional Sinkhorn scaling equations are analytic with a uniform implicit-function radius on $\mathcal{B}$. The debiased Sinkhorn divergence therefore admits a uniform first-order expansion with quadratic remainder in the weight perturbation.

The empirical weight vector is a sum of bounded independent categorical vectors on at most m_0 atoms, so

$$\mathbb{E}\,\|\widehat{b}-b\|_1 \le \sqrt{m_0\,\mathbb{E}\,\|\widehat{b}-b\|_2^2} = O(n^{-1/2}).$$

Indeed, each coordinate is an average of bounded independent indicators, so the sum of coordinate variances is $O(1/n)$ uniformly on the admissible class. The induced influence class is therefore uniformly root-n on the admissible finite-support class. Combining the uniform linearization with the abstract horizon law at $a = 1/2$ yields the stated bound and horizon scale.

Proposition 2.17 (Pointwise fixed-ε inheritance in the 1D proof model)**.** *Assume the hypotheses of proposition 2.7, let $\widehat{P}_n^{\mathrm{iid}}$ be the empirical measure of n i.i.d. samples from $\bar{P}_n$, and fix $\varepsilon > 0$. For the debiased Sinkhorn divergence S_ε with squared-Euclidean cost on the common compact support, one has*

$$\mathbb{E}\,S_\varepsilon\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) = O(n^{-1/2}).$$

Proof. Write $V_\varepsilon(\mu,\nu)$ for the regularized OT value so that

$$S_\varepsilon(\mu,\nu) = V_\varepsilon(\mu,\nu) - \tfrac{1}{2}V_\varepsilon(\mu,\mu) - \tfrac{1}{2}V_\varepsilon(\nu,\nu).$$

Because all measures are supported on one compact interval, they lie in a common bounded quadratic-moment class. By Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4], for fixed ε there exists $L_\varepsilon < \infty$ depending only on that class and on the cost such that

$$|V_\varepsilon(\mu,\nu) - V_\varepsilon(\mu',\nu')| \le L_\varepsilon\big(W_2(\mu,\mu') + W_2(\nu,\nu')\big).$$

Applying this bound to the cross term and both self terms gives

$$|S_\varepsilon(\mu,\nu) - S_\varepsilon(\mu',\nu')| \le 2L_\varepsilon\big(W_2(\mu,\mu') + W_2(\nu,\nu')\big).$$

With $\nu = \nu' = \bar{P}_n$,

$$S_\varepsilon\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) \le S_\varepsilon\Big(\widehat{P}_n^{\mathrm{iid}}, \bar{P}_n\Big) + 2L_\varepsilon W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \widehat{P}_n^{\mathrm{iid}}\Big).$$

Taking expectations and using the triangle inequality gives

$$\mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \widehat{P}_n^{\mathrm{iid}}\Big) \le \mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) + \mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{iid}}, \bar{P}_n\Big).$$

The first term is $O(n^{-1/2})$ by proposition 2.7. The second is the classical i.i.d. one-dimensional quantile-process rate on the same compact support class [Bahadur, 1966, van der Vaart, 1998], so it is also $O(n^{-1/2})$. Hence

$$\mathbb{E}\,W_2\Big(\widehat{P}_n^{\mathrm{tri}}, \widehat{P}_n^{\mathrm{iid}}\Big) = O(n^{-1/2}).$$

Finally, under the null comparison with fixed ε and compact support, the i.i.d. debiased Sinkhorn term satisfies

$$\mathbb{E}\,S_\varepsilon\Big(\widehat{P}_n^{\mathrm{iid}}, \bar{P}_n\Big) = O(n^{-1})$$

by del Barrio et al. [2023, Thm. 5.1, after fixed-ε rescaling]; see also the null-limit law in Goldfeld et al. [2024]. Combining the bounds yields the displayed $O(n^{-1/2})$ rate.

Proposition 2.18 (Bandwise fixed-ε inheritance in the 1D proof model)**.** *Let $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}] \subset (0,\infty)$ be a compact regularization band. Under the hypotheses of proposition 2.17 on a common compact support class, there exists $C_{\mathcal{B}} < \infty$ such that*

$$\sup_{\varepsilon\in\mathcal{B}} \mathbb{E}\,S_\varepsilon\Big(\widehat{P}_n^{\mathrm{tri}}, \bar{P}_n\Big) \le C_{\mathcal{B}} n^{-1/2}.$$

Consequently, the induced horizon scale is uniform on $\mathcal{B}$:

$$\sup_{\varepsilon\in\mathcal{B}} n_\varepsilon^*(H) \asymp (C_{\mathcal{B}}/\zeta)^{1/(H+1/2)}.$$

Proof. Repeat the proof of proposition 2.17. The stability constant L_ε from Eckstein and Nutz [2022, Thm. 3.7(ii), Ex. 3.4] is bounded on $\mathcal{B}$ because all measures live in one compact support class and $\varepsilon \ge \varepsilon_{\min} > 0$. The i.i.d. debiased-Sinkhorn benchmark on compact support is also uniform on $\mathcal{B}$ by the fixed-ε asymptotics of Goldfeld et al. [2024] and the compact-support null bounds of del Barrio et al. [2023, Thm. 5.1, after fixed-ε rescaling]. Since the triangular-to-i.i.d. W_2 comparison is independent of ε, the constants in the pointwise proof can be chosen uniformly over $\mathcal{B}$.

Lemma 2.19 (Exact ℓ_2 fluctuation on the calibrated support-growth grid)**.** *In the calibrated discrete $k = 2$ fixed-span grid with span 0.25, let $b_{t,n} \in \mathbb{R}^{4n}$ be the weight vector of $\bar{P}_t^{(n)}$ and let $\widehat{b}_{t,n}$ be the corresponding triangular-window empirical weight vector. Let $\{v_{r,\ell} : 1 \le r \le n,\ 1 \le \ell \le 3\}$ be any orthonormal block-contrast basis of the tangent space obtained by placing an orthonormal contrast basis on each four-point block. Then*

$$\mathbb{E}\,\langle \widehat{b}_{t,n} - b_{t,n}, v_{r,\ell}\rangle\langle \widehat{b}_{t,n} - b_{t,n}, v_{s,m}\rangle = \frac{1}{4n^2}\mathbf{1}\{(r,\ell) = (s,m)\}.$$

In particular,

$$\mathbb{E}\,\|\widehat{b}_{t,n} - b_{t,n}\|_2^2 = \frac{3}{4n}.$$

If $\widehat{b}_{t,n}^{\mathrm{iid}}$ is the i.i.d. empirical weight vector from n samples drawn from the same $4n$-atom uniform mixture, then

$$\mathbb{E}\,\|\widehat{b}_{t,n}^{\mathrm{iid}} - b_{t,n}\|_2^2 = \frac{1-1/(4n)}{n}.$$

Proof. The calibrated support-growth grid splits into n disjoint four-point blocks, each carrying target mass $1/n$ and uniform within-block weights $1/4$. The triangular sample chooses exactly one atom from each block. On one block, the weight error vector is $(e_J - u/4)/n$, where J is uniform on $\{1,2,3,4\}$ and $u = (1,1,1,1)$. Therefore

$$\mathbb{E}\,\|e_J - u/4\|_2^2 = \left(\frac{3}{4}\right)^2 + 3\left(\frac{1}{4}\right)^2 = \frac{3}{4}.$$

Summing over the n disjoint blocks gives the block-basis covariance identity, and therefore

$$\mathbb{E}\,\|\widehat{b}_{t,n} - b_{t,n}\|_2^2 = n\cdot\frac{3}{4n^2} = \frac{3}{4n}.$$

For the i.i.d. empirical vector on a uniform $4n$-atom support,

$$\mathbb{E}\,\|\widehat{b}^{\mathrm{iid}}_{t,n} - b_{t,n}\|_2^2 = \frac{1-\sum_{i=1}^{4n} b_i^2}{n} = \frac{1-1/(4n)}{n}.$$

Lemma 2.20 (Parity decomposition of the calibrated row kernel)**.** *In the calibrated discrete $k=2$ fixed-span grid, write each row support as*

$$x_{r,\sigma_x,\sigma_y} = \big(s_r + \sigma_x\rho,\ \sigma_y\rho\big), \qquad \sigma_x,\sigma_y \in \{-1,1\},$$

with row shift $s_r \in [0, 0.25]$ and radius $\rho = 0.125$. For $\varepsilon > 0$, let $K_{rs}^{(\varepsilon)}$ be the 4×4 Sinkhorn kernel block between rows r and s. Then, with $\Delta_{rs} = s_r - s_s$,

$$K_{rs}^{(\varepsilon)} = K_x^{(\varepsilon)}(\Delta_{rs}) \otimes K_y^{(\varepsilon)},$$

where

$$K_x^{(\varepsilon)}(\Delta) = \begin{bmatrix} e^{-\Delta^2/\varepsilon} & e^{-(\Delta+2\rho)^2/\varepsilon} \\ e^{-(\Delta-2\rho)^2/\varepsilon} & e^{-\Delta^2/\varepsilon} \end{bmatrix}$$

and

$$K_y^{(\varepsilon)} = \begin{bmatrix} 1 & e^{-4\rho^2/\varepsilon} \\ e^{-4\rho^2/\varepsilon} & 1 \end{bmatrix}.$$

Let

$$H = 2^{-1/2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad W = H\otimes H.$$

Then

$$WK_{rs}^{(\varepsilon)}W^\top = \widetilde{K}_x^{(\varepsilon)}(\Delta_{rs}) \otimes \mathrm{diag}\big(1+e^{-4\rho^2/\varepsilon},\, 1-e^{-4\rho^2/\varepsilon}\big),$$

where $\widetilde{K}_x^{(\varepsilon)}(\Delta) = HK_x^{(\varepsilon)}(\Delta)H^\top$. Consequently the row-wise four-point space splits exactly into a one-dimensional odd channel and a two-dimensional even channel, and the calibrated tangent space inherits the same invariant decomposition.

Proof. The squared-Euclidean cost separates by coordinate:

$$\|x_{r,\sigma_x,\sigma_y} - x_{s,\tau_x,\tau_y}\|^2 = (\Delta_{rs} + \rho(\sigma_x - \tau_x))^2 + \rho^2(\sigma_y - \tau_y)^2.$$

Exponentiating the negative cost divided by ε therefore gives the stated tensor factorization of the kernel block. The matrix H diagonalizes the two-point symmetric kernel in the y coordinate, and tensoring with the same transform in the x coordinate yields the displayed formula. The final claim follows because the transformed block matrix is block diagonal with respect to the y-parity splitting.

Lemma 2.21 (Quadratic null bound from a uniform Hessian bound)**.** *For the calibrated support-growth family, fix (t,n) and let $F_{t,n,\varepsilon}(h)$ denote the null Sinkhorn map*

$$F_{t,n,\varepsilon}(h) := S_\varepsilon\left(\sum_{i=1}^{4n}(b_{t,n,i}+h_i)\delta_{x_i},\ \sum_{i=1}^{4n} b_{t,n,i}\delta_{x_i}\right)$$

on the tangent domain

$$\mathcal{T}_{t,n} := \{h\in\mathbb{R}^{4n} : \sum_{i=1}^{4n} h_i = 0,\ b_{t,n}+h\in\Delta_{4n}\}.$$

Assume that for some finite $M_{\mathcal{B}}$ the restriction of $F_{t,n,\varepsilon}$ to $\mathcal{T}_{t,n}$ is twice continuously differentiable for every $\varepsilon\in\mathcal{B}$, and that

$$\sup_{\varepsilon\in\mathcal{B}}\sup_{(t,n)}\sup_{h\in\mathcal{T}_{t,n}}\big\|D^2F_{t,n,\varepsilon}(h)\big\|_{2\to 2} \le M_{\mathcal{B}}.$$

Then

$$\sup_{\varepsilon\in\mathcal{B}} F_{t,n,\varepsilon}(h) \le \frac{M_{\mathcal{B}}}{2}\|h\|_2^2 \qquad \text{for every } h\in\mathcal{T}_{t,n}.$$

In particular, the hypothesis of theorem 2.27 holds with $H_{\mathcal{B}} = M_{\mathcal{B}}/2$.

Proof. For each fixed (t,n,ε), the debiased Sinkhorn divergence is nonnegative and vanishes at the null comparison, so $F_{t,n,\varepsilon}(0)=0$ and 0 is a minimizer on $\mathcal{T}_{t,n}$. Since $F_{t,n,\varepsilon}$ is C^2 on the tangent domain, its first derivative on that domain vanishes at 0. For any $h\in\mathcal{T}_{t,n}$, the one-variable path $g(s) = F_{t,n,\varepsilon}(sh)$ satisfies $g(0)=g'(0)=0$, and Taylor's formula gives

$$F_{t,n,\varepsilon}(h) = \frac{1}{2}g''(\xi) = \frac{1}{2}\langle h, D^2F_{t,n,\varepsilon}(\xi h)h\rangle$$

for some $\xi\in(0,1)$. Therefore

$$F_{t,n,\varepsilon}(h) \le \frac{1}{2}\big\|D^2F_{t,n,\varepsilon}(\xi h)\big\|_{2\to 2}\|h\|_2^2 \le \frac{M_{\mathcal{B}}}{2}\|h\|_2^2.$$

Taking the supremum over $\varepsilon\in\mathcal{B}$ yields the claim.

Remark 2.22 (Parity-channel curvature decomposition). In the block-contrast basis, local directional curvatures remain $O(1)$ across the reported grid, whereas collective directional curvatures grow approximately like $c(\varepsilon)n$ with $c(\varepsilon)$ stable across n. The exact isotropic covariance of order $1/n^2$ in the same block basis translates a linear trace bound into an $O(n^{-1})$ expected null cost.

Proposition 2.23 (Covariance-Weighted Null Bound for Calibrated Support Growth)**.** *For the calibrated support-growth family, let $H_{t,n,\varepsilon}$ denote the null Hessian of the debiased Sinkhorn map in the orthonormal block-contrast basis. Suppose that for some $G_{\mathcal{B}}<\infty$ one has*

$$\sup_{\varepsilon\in\mathcal{B}}\sup_{(t,n)}\mathrm{tr}\big(H_{t,n,\varepsilon}\big) \le G_{\mathcal{B}}n$$

and that the null map is twice differentiable with a nonnegative quadratic remainder along the tangent domain. Then

$$\sup_{\varepsilon\in\mathcal{B}}\mathbb{E}\,S_\varepsilon\Big(\widehat{P}^{(n)}_{t,\mathrm{tri}}, \bar{P}^{(n)}_t\Big) \le \frac{G_{\mathcal{B}}}{8n} + o(n^{-1}).$$

Consequently the support-growth family has finite-sample exponent ‘a=1‘ under this trace condition.

Proof. Work in the orthonormal block-contrast basis. By the exact covariance identity in lemma 2.19, the empirical tangent covariance is $(4n^2)^{-1}I$. The second-order Taylor expansion of the null map therefore gives

$$\mathbb{E}\, S_\varepsilon \leq \frac{1}{2}\operatorname{tr}\left(H_{t,n,\varepsilon}\,\frac{1}{4n^2}I\right)+o(n^{-1})=\frac{\operatorname{tr}(H_{t,n,\varepsilon})}{8n^2}+o(n^{-1}).$$

The trace bound then yields the displayed $O(n^{-1})$ rate.

Corollary 2.24 (Uniform Local Block-Curvature Bound for Support Growth)**.** *In the calibrated support-growth family, let $\{v_{r,\ell} : 1 \leq r \leq n,\ 1 \leq \ell \leq 3\}$ be the orthonormal block- contrast basis from lemma 2.19. Suppose that for some $L_{\mathcal{B}} < \infty$ one has*

$$\sup_{\varepsilon\in\mathcal{B}}\sup_{(t,n)}\sup_{1\leq r\leq n}\sup_{1\leq \ell\leq 3}\langle v_{r,\ell}, H_{t,n,\varepsilon}v_{r,\ell}\rangle \leq L_{\mathcal{B}}.$$

Then

$$\sup_{\varepsilon\in\mathcal{B}}\mathbb{E}\, S_\varepsilon\left(\widehat{P}^{(n)}_{t,\mathrm{tri}},\bar{P}^{(n)}_t\right)\leq \frac{3L_{\mathcal{B}}}{8n}+o(n^{-1}).$$

In particular, the calibrated support-growth family has finite-sample exponent 'a=1' under a uniform local block-curvature bound.

Proof. In the orthonormal block-contrast basis,

$$\operatorname{tr}(H_{t,n,\varepsilon})=\sum_{r=1}^{n}\sum_{\ell=1}^{3}\langle v_{r,\ell},H_{t,n,\varepsilon}v_{r,\ell}\rangle \leq 3L_{\mathcal{B}}n.$$

Apply proposition 2.23 with $G_{\mathcal{B}} = 3L_{\mathcal{B}}$.

Remark 2.25 (Channel reduction)*.* By lemma 2.20, the calibrated row kernel has only two nontrivial parity sectors: a one-dimensional odd channel and a two-dimensional even channel. The local block-curvature criterion in corollary 2.24 controls the reduced channels on the full '3n'-dimensional tangent space.

Proposition 2.26 (Uniform Local Parity-Channel Curvature Bound)**.** *For the calibrated support-growth family, write the row position as*

$$u_r := \frac{r-1}{n-1}\in[0,1],$$

and let $\kappa_1(u,\varepsilon)$, $\kappa_2(u,\varepsilon)$, and $\kappa_3(u,\varepsilon)$ denote the second derivatives at the null comparison of the debiased Sinkhorn map in the odd channel and the two even channels obtained from lemma 2.20. Then each κ_j is continuous on $\mathcal{K} := [0,1]\times\mathcal{B}$, hence there exists a finite constant $L_{\mathcal{B}}$ such that

$$\sup_{(u,\varepsilon)\in\mathcal{K}}\max_{1\leq j\leq 3}\kappa_j(u,\varepsilon)\leq L_{\mathcal{B}}.$$

Consequently the local block-curvature criterion in corollary 2.24 holds uniformly on the calibrated support-growth family.

Proof. The calibrated row kernel depends analytically on (u,ε), and the kernel blocks are strictly positive on the compact set $\mathcal{K}$. The finite- dimensional Sinkhorn scaling equations therefore admit an analytic solution in the neighborhood of the null comparison by the implicit function theorem. The debiased Sinkhorn map is a composition of analytic maps in the weights and the kernel blocks, so its channelwise second derivatives are continuous on $\mathcal{K}$. A continuous function on a compact set attains its maximum, which gives the displayed uniform bound.

Theorem 2.27 (Calibrated Support-Growth Sinkhorn Horizon Law)**.** *Consider the calibrated discrete $k = 2$ fixed-span support-growth grid and let $b_{t,n}$ denote the target weight vector on the common $4n$-point support. Fix a compact band $\mathcal{B} = [\varepsilon_{\min}, \varepsilon_{\max}]$. Then the induced temporal-validity horizon on this support-growth family has exponent 'a=1':*

$$n^*_{\mathcal{B}}(1,H)\asymp(G_{\mathcal{B}}/\zeta)^{1/(H+1)}.$$

Proof. Combine proposition 2.23 and proposition 2.26 with the abstract horizon law at finite-sample exponent 'a=1'.

The theorem-level line closes on the bounded-cardinality and calibrated support-growth families. Support-changing and singular embedded Sinkhorn classes remain outside the proved regime.

3 Transfer and Operational Implications

Finite projection families and sliced geometries already transfer the horizon law under uniform projected control, and the embedded Sinkhorn results in the main theory cover the bounded-cardinality finite-discrete and calibrated support-growth families. High-dimensional practicality then becomes a geometric question: temporal validity remains usable only when the chosen geometry preserves a usable finite-sample exponent.

Corollary 3.1 (Effective-exponent horizon transfer)**.** *Let d_{eff} be a probability metric such that, on the drifting class of interest,*

$$\mathbb{E}\, d_{\mathrm{eff}}\left(\widehat{P}^{(n)}_t,\bar{P}^{(n)}_t\right)\leq C_{K,\mathrm{eff}}n^{-a_{\mathrm{eff}}}$$

for some $a_{\mathrm{eff}} > 0$, and suppose the same class satisfies the staleness bound

$$d_{\mathrm{eff}}\left(\bar{P}^{(n)}_t,P_t\right)\leq C_{S,\mathrm{eff}}\zeta n^H.$$

Then

$$\mathbb{E}\, d_{\mathrm{eff}}\left(\widehat{P}^{(n)}_t,P_t\right)\leq C_{K,\mathrm{eff}}n^{-a_{\mathrm{eff}}}+C_{S,\mathrm{eff}}\zeta n^H,$$

and the induced temporal-validity horizon satisfies

$$n^*_{\mathrm{eff}}(H)\asymp(C_{K,\mathrm{eff}}/\zeta)^{1/(a_{\mathrm{eff}}+H)}.$$

Proof. Apply Theorem 2.1 with $d = d_{\mathrm{eff}}$.

The practical question is therefore not high dimension in itself, but which geometry preserves a usable exponent a_{eff}. Finite projection and sliced constructions target root-n one-dimensional behavior. Intrinsic- dimension and projection-robust transport can replace ambient dimension by an effective dimension. Smoothed transport can preserve parametric behavior through regularization [Weed and Bach, 2019, Paty and Cuturi, 2019, Nietert et al., 2022]. Geometry selection is therefore memory-scale selection. Weighted-window diagnostics on uniform, triangular, and geometric tapers show the same finite-sample versus lag trade-off in operational form; see Table 11.

Operational use begins with horizon inference: the plug-in horizon must enter the useful-memory region and remain there under streaming noise.

Support change beyond the fixed support-growth construction and detector-uniform online control remain open.

4 Horizon Inference

Operational use requires more than the existence of a horizon. A data-driven memory rule needs estimators of the drift parameters (H, ζ) together with a criterion that compares inferential resolution to the curvature of the useful-memory region.

Write $\alpha = \log \zeta$, and let m denote the per-lag sample size used to estimate lagged transport discrepancies. For a fixed lag design $1, \dots, L$, define the lag-energy information scale

$$\mathcal{I}_{m,L}(H) \coloneqq m \sum_{j=1}^{L} j^{2H}.$$

Assumption 4.1 (Lag-geometry inference input)**.** For the lag model $D_j = \zeta j^H$, suppose there are estimators $(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ such that, on compact parameter sets,

$$\sqrt{\mathcal{I}_{m,L}(H)} \begin{pmatrix} \widehat{\alpha}_{m,L} - \alpha \\ \widehat{H}_{m,L} - H \end{pmatrix} \Rightarrow \mathcal{N}(\mathbf{0}, \Sigma_{L,H,\zeta}).$$

The power-law lag-geometry model of Parreira [2026b] provides one concrete source of this assumption.

The continuous optimizer of the horizon envelope can be written as the smooth map

$$n_\star(\alpha, H) = \left(\frac{a C_K}{H C_S e^\alpha} \right)^{1/(a+H)},$$
$$g(\alpha, H) \coloneqq \log n_\star(\alpha, H).$$

Its gradient is

$$\nabla g(\alpha, H) = \begin{pmatrix} -(a+H)^{-1} \\ -\left[H(a+H)\right]^{-1} - g(\alpha, H)(a+H)^{-1} \end{pmatrix}.$$

Theorem 4.2 (Plug-in Horizon Central Limit Law)**.** *Under Assumption 4.1,*

$$\sqrt{\mathcal{I}_{m,L}(H)} \big(\log \widehat{n}_\star - \log n_\star\big) \Rightarrow \mathcal{N}(0, \tau^2),$$

where $\widehat{n}_\star = n_\star(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ *and*

$$\tau^2 = \nabla g(\alpha, H)^\top \Sigma_{L,H,\zeta} \nabla g(\alpha, H).$$

Proof. The map g is smooth on any compact set bounded away from $H = 0$ and $\zeta = 0$. Apply the multivariate delta method to $(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ under Assumption 4.1.

The useful-memory region is the set

$$\mathcal{U}_\delta = \{n : \Phi(n) \le (1+\delta)\Phi(n_\star)\}.$$

Let $(x_-(\delta; a, H), x_+(\delta; a, H))$ be the normalized endpoints from Proposition 2.5, and define the log-radius

$$r_\delta(a, H) \coloneqq \min\big\{-\log x_-(\delta; a, H), \log x_+(\delta; a, H)\big\}.$$

This quantity is the largest symmetric log-error around $n_\star$ that stays inside the useful-memory region.

Theorem 4.3 (Operational Identifiability of Useful Memory)**.** *Under Assumption 4.1,*

$$\mathbb{P}\big(\widehat{n}_\star \in \mathcal{U}_\delta\big) = 2\Phi_{\mathcal{N}}\left(\frac{\sqrt{\mathcal{I}_{m,L}(H)}\, r_\delta(a, H)}{\tau} \right) - 1 + o(1),$$

where $\Phi_{\mathcal{N}}$ *is the standard normal CDF. In particular,*

$$\frac{\sqrt{\mathcal{I}_{m,L}(H)}\, r_\delta(a, H)}{\tau(H, \zeta)} \to \infty \quad \Longrightarrow \quad \mathbb{P}\big(\widehat{n}_\star \in \mathcal{U}_\delta\big) \to 1.$$

Proof. By Proposition 2.5, $\widehat{n}_\star \in \mathcal{U}_\delta$ if and only if $|\log \widehat{n}_\star - \log n_\star| \le r_\delta(a, H)$. Standardize this event with Theorem 4.2 and use the symmetry of the Gaussian limit.

The quantity

$$\mathfrak{S}_{\delta,m,L} \coloneqq \frac{\sqrt{\mathcal{I}_{m,L}(H)}\, r_\delta(a, H)}{\tau(H, \zeta)}$$

is therefore the operational identifiability score of the horizon. It compares lag-geometry information to the curvature-imposed precision requirement of the useful-memory region.

Proposition 4.4 (Deterministic stability of the horizon map)**.** *Let* $K \subset \mathbb{R} \times (0, \infty)$ *be compact. Then there exists a finite constant* L_K *such that, for all* $(\alpha_1, H_1), (\alpha_2, H_2) \in K$*,*

$$|g(\alpha_1, H_1) - g(\alpha_2, H_2)| \le L_K \big(|\alpha_1 - \alpha_2| + |H_1 - H_2|\big).$$

Consequently, if (α, H) *and* $(\widehat{\alpha}_{m,L}, \widehat{H}_{m,L})$ *lie in* K *and*

$$|\widehat{\alpha}_{m,L} - \alpha| + |\widehat{H}_{m,L} - H| \le \frac{r_\delta(a, H)}{L_K},$$

then $\widehat{n}_\star \in \mathcal{U}_\delta$*.*

Proof. The gradient of g is continuous on K, so $\|\nabla g\|_1$ is bounded there. Apply the mean-value theorem to obtain the Lipschitz bound. The inclusion claim follows from $|\log \widehat{n}_\star - \log n_\star| \le r_\delta(a, H)$ together with the characterization of $\mathcal{U}_\delta$ from Proposition 2.5.

The informative regime lies across the transition from low to moderate identifiability. Table Table 1 shows representative points across that transition.

Table 1: Representative regimes across the inferable-horizon transition.

L	m	H	δ	$\mathfrak{S}$	Empirical	Gaussian
5	20	0.3	0.050	0.501	0.411	0.384
8	10	0.9	0.020	1.001	0.633	0.683
3	200	0.3	0.010	1.996	0.956	0.954
8	100	0.3	0.005	3.001	0.996	0.997

On the tested grid, the plug-in central limit law is also well calibrated at the scale level: the empirical standard deviation of $\log \widehat{n}_\star$ tracks the delta-method prediction closely, and the nominal 95% Gaussian interval coverage stays near its target across the reported regimes. The simulation suite also calibrates the regret expansion directly.

Proposition 4.5 (Deterministic and asymptotic plug-in horizon regret)**.** *Let* $u = \log(\widehat{n}_\star / n_\star)$. *Then, as* $u \to 0$,

$$\frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} = \Psi(e^u) - 1 = \frac{He^{-au} + ae^{Hu}}{a+H} - 1.$$

Consequently, for every $r > 0$,

$$|u| \le r \quad \Longrightarrow \quad \frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} \le B_r(a, H),$$

where

$$B_r(a, H) := \max\{\Psi(e^r) - 1,\ \Psi(e^{-r}) - 1\}.$$

Moreover,

$$\frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)} = \frac{aH}{2} u^2 + O(u^3).$$

Consequently,

$$\mathbb{E}\left[\frac{\Phi(\widehat{n}_\star) - \Phi(n_\star)}{\Phi(n_\star)}\right] = \frac{aH}{2} \frac{\tau^2}{\mathcal{I}_{m,L}(H)} + o\big(\mathcal{I}_{m,L}(H)^{-1}\big)\,.$$

Proof. By Proposition 2.5,

$$\frac{\Phi(n_\star e^u)}{\Phi(n_\star)} = \Psi(e^u) = \frac{He^{-au} + ae^{Hu}}{a+H}.$$

The deterministic bound follows by maximizing this exact expression over $u \in [-r, r]$. Differentiate at $u = 0$. The first derivative vanishes and the second derivative equals aH, giving the quadratic expansion. The expectation bound then follows from Theorem 4.2 and the boundedness of third moments on compact parameter sets.

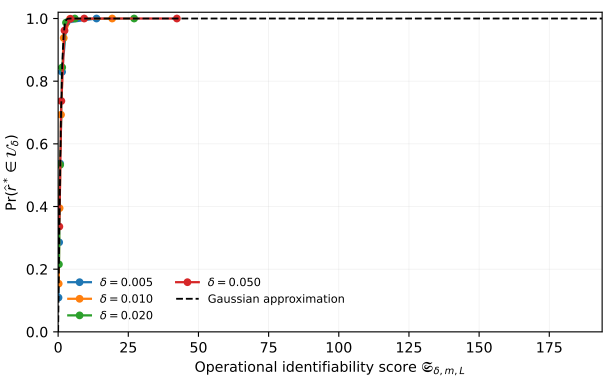

Figure 3: Operational identifiability of the useful-memory region. Binning by the operational score reveals a monotone rise in useful-region hit rate, with each δ curve tracking the Gaussian approximation from Theorem 4.3.

Structural existence and operational identifiability need not coincide. The relevant scale is the score $\mathfrak{S}_{\delta,m,L}$, which organizes the transition from low coverage to near-certain useful-memory recovery.

5 Empirical Signatures

The experiments target qualitative signatures of the temporal-validity horizon: an interior memory optimum, roughness-dependent scaling, the same pattern on a real stream, structured support for the fixed-ε Sinkhorn family, and validity loss before detector alarm. The reported sweeps use repeated Monte Carlo runs and are meant to identify shape, ordering, and sign rather than to estimate constants precisely.

5.1 Interior Memory Optimum

The Gaussian horizon sweep isolates the variance–staleness trade-off directly. Small windows are variance dominated, large windows are stale, and the minimum lies at an interior horizon.

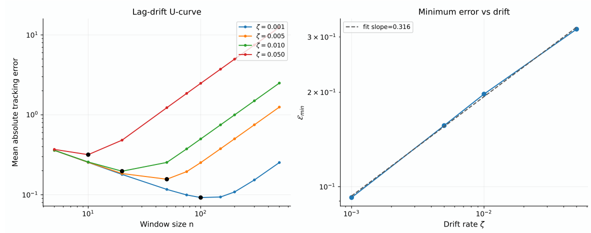

Figure 4: Interior risk optimum over memory length. Tracking error first falls with horizon and then rises once staleness overtakes the remaining variance gain. The optimum remains finite across drift strengths.

This interior optimum is the basic empirical signature of temporal validity under drift. Under stationarity, more memory would continue to help. Under drift, the horizon that reduces variance also ages the evidence. Around the minimizing horizon, the curve displays a flat near-optimal region, which is the empirical analogue of the useful-memory region defined from the envelope Φ.

5.2 Roughness-Dependent Horizon Scaling

The roughness-family experiment varies the roughness index directly. For each Hölder-type regime, the surrogate error obeys the same structural decomposition as the theory,

$$\mathcal{E}(n) \approx \sigma n^{-1/2} + \zeta n^H,$$

and the experiment traces the empirically optimal horizon across a sweep of ζ values.

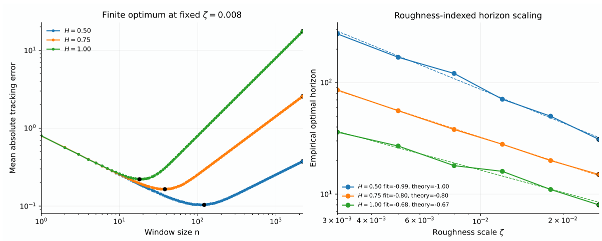

Figure 5: Roughness-dependent horizon scaling. Each path class exhibits a finite optimum, and the location of that optimum changes with H. The horizon is a function of drift regularity, not a fixed window rule.

The empirical optima separate by roughness class, matching the path-dependent horizon logic of corollary 2.4.

5.3 Real-Stream Memory Diagnostic

A similar signature appears on a real nonstationary stream. On ELEC2 [Harries, 1999], the diagnostic compares a short anchor block with the preceding memory window in W_1, yielding a real-data analogue of the finite-memory U-curve.

The optimum occurs at a finite memory scale with a short near-optimal band. The same retention pattern therefore appears on a real stream: very short windows are too noisy, while very long windows become stale relative to

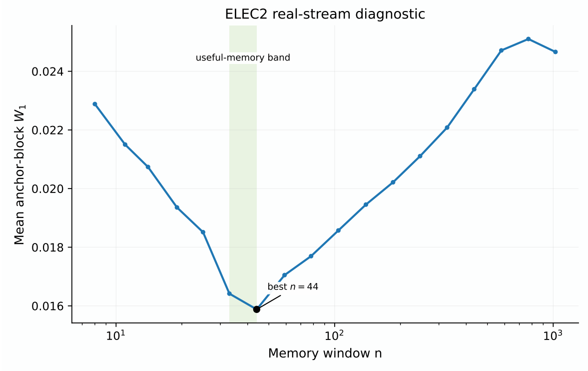

Figure 6: ELEC2 real-stream diagnostic on the `nswprice` marginal. The mean anchor-block W_1 error exhibits a finite interior optimum and a near-optimal useful-memory band, matching the temporal-validity signature on a real stream.

the present block. On the robustness grid, the `nswprice` marginal retains an interior optimum in all tested configurations, and the `nswdemand` marginal shows the same interior pattern with more movement in the optimizing scale; see Table 8.

5.4 Moderate-Band Sinkhorn Diagnostics

The fixed-ε Sinkhorn experiments ask whether a practical geometry can keep the finite-sample term stable enough for the horizon law to remain informative in an embedded regime. On the tested low-intrinsic-dimensional families, the finite-sample term stays close to the i.i.d. benchmark across a moderate regularization band, with the widest stable regimes on $(8,1)$, $(8,2)$, and $(12,2)$ and a narrower regime on $(12,1)$. Direct probes localize the remaining instability in the self-coupling blocks, while the calibrated certificate identifies a moderate band extending beyond the bounded-cardinality finite-support theorem.

On the calibrated discrete $k = 2$ support grid, the theorem-facing diagnostics support the same picture: self-coupling remains contractive, exact-target null slopes stay near the quadratic scale, finite-support fluctuation proxies remain bounded, and the support-growth cost-to-ℓ_2^2 ratio stays stable across the reported band. Together with the exact ℓ_2 fluctuation identity, these diagnostics organize the support-growth family around a covariance-weighted trace bound and a local block-curvature bound for the null Sinkhorn map. The finite-difference Hessian probe shows bounded local curvatures together with collective directions that grow approximately linearly with support growth, matching the parity-channel reduction used in the theorem line.

Intrinsic-dimension estimates from TwoNN [Facco et al., 2017] track the stable-band cutoff better than ambient dimension on the tested grid, so the regime-selection variable is local geometry rather than ambient dimension. Detailed calibration tables appear in Appendix B.

5.5 Validity Loss Before Detection

Temporal validity and changepoint evidence are different clocks. To make that separation explicit, consider a stream whose local drift rate ramps upward over time. The horizon n_t^* contracts as the ramp steepens, while a long operating horizon remains fixed. Define

$$\Delta_{\mathrm{val-det}} := t_{\mathrm{detect}} - t_{\mathrm{valid}},$$

where t_{valid} is the first time the operating horizon persistently exceeds the temporal-validity horizon, and t_{detect} is the first detector alarm that follows.

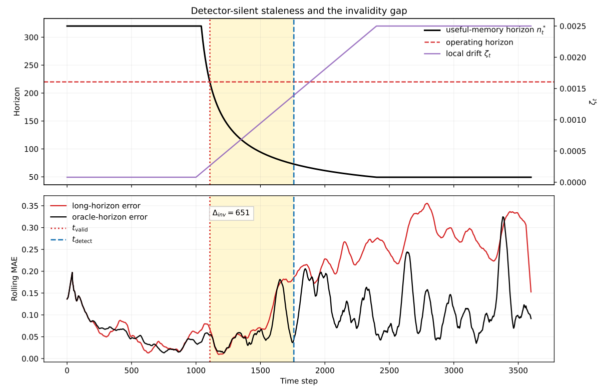

Figure 7: Temporal validity can fail before detector alarm. As drift intensifies, the temporal-validity horizon contracts until the operating horizon becomes stale at t_{valid}; the detector responds only later at t_{detect}, leaving a strictly positive validity-detection lag. The lower panel shows excess tracking error accumulating before the alarm appears.

Across the reported runs, the measured lag remains positive. Retained evidence can already be invalid for the present while detector statistics remain below alarm threshold.

The positive lag is not a knife-edge feature of one tuning choice. Across the tested ADWIN thresholds [Bifet and Gavaldà, 2007], mean detection time remains strictly later than mean validity-expiry time, and the qualitative lag persists across detector families and input representations on the tested grid. On the raw-observation input, the full tested grid of ADWIN and Page–Hinkley detectors [Page, 1954, Hinkley, 1970], operating windows 160, 220, and 300, and the reported threshold sweep keeps positive-gap rate equal to one together with unit detection rate. Page–Hinkley reacts earlier on this drift family, with mean gaps between about 205 and 345, while ADWIN yields mean gaps between about 429 and 620. Residualized detector inputs are not uniformly better on this stream: ADWIN on absolute residuals fails to trigger on the tested grid, whereas Page–Hinkley still detects but with lower detection rates and substantially larger delays. The empirical claim is robustness of the sign of $\Delta_{\mathrm{val-det}}$ on the tested observation-based detector/tuning family, not a detector-uniform delay law; see Table 9.

5.6 Empirical Summary

The experiments support a structural U-curve with a near-optimal useful-memory region, roughness-dependent horizon scaling, a corresponding pattern on the ELEC2 stream, structured support for the fixed-ε Sinkhorn family, and a measurable lag between validity loss and detector alarm. Complementary online-adaptation sweeps across default, smooth, rough, alternating, and ramp-up profiles show that the structural controller keeps mean error between 1.18 and 1.36 times oracle risk, versus 1.34–1.65 for the activity baseline and 2.30–4.41 for the raw plug-in rule; see Table 10.

6 Related Work

Drifted optimization and regret. Dynamic-regret and variation-budget analyses show that under drift, forgetting scales and window lengths must adapt to how quickly the target moves [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023, Wang et al., 2024]. The object studied here is earlier in the pipeline: it asks how long retained evidence remains temporally valid before age becomes the dominant source of tracking error.

The same structural trade-off also appears in drifting density estimation. Mazzetto and Upfal [2023] derive minimax rates by optimizing over recent-memory windows under bounded drift, which places the present horizon law in direct contact with finite-memory risk optimization under nonstationarity.

Adaptive windows and detection. Adaptive-window and concept-drift methods study when accumulated evidence is sufficient to react [Bifet and Gavaldà, 2007, Gama et al., 2004, 2014]. Those works provide the reaction timescale. The horizon law adds the validity timescale: retained evidence can cease to represent the present before change statistics cross an alarm threshold.

Sequential nonparametric testing. Sequential two-sample testing and kernel change detection quantify the evidence needed to reject stability under time-uniform false-alarm control. These procedures operate on a detection timescale. The temporal-validity horizon identifies the tracking timescale on which retained evidence ceases to represent the present, so the two clocks need not agree. Exponentially weighted MMD detectors make this memory dependence explicit by replacing a single fixed window with implicit forgetting scales while still calibrating a detection threshold [Kalinke et al., 2025].

Filtering and local smoothing. Classical adaptive filtering and state-space tracking tune forgetting factors, gain schedules, or process-noise levels to balance smoothing against responsiveness [Kalman, 1960, Jazwinski, 1970]. Locally stationary smoothing and bandwidth selection solve the analogous bias–variance problem for evolving regression or spectral targets [Dahlhaus, 1997, Vogt, 2012]. Those literatures make the same trade-off visible as a variance term versus a drift-induced local bias term. The horizon law lifts that logic from scalar smoothing targets to distribution tracking, where the finite-sample exponent depends on the chosen probability geometry.

Transport geometry. Wasserstein distance and Sinkhorn-type estimators provide the finite-sample geometries that enter the horizon law [Fournier and Guillin, 2015, Boissard and Le Gouic, 2014, Weed and Bach, 2019, Genevay et al., 2019, Mena and Niles-Weed, 2019, del Barrio et al., 2023, Goldfeld et al., 2024]. Sliced and projection-robust variants show that geometry choice can change the finite-sample exponent, while intrinsic dimension and regularization calibration determine which regime is available [Kolouri et al., 2018, Deshpande et al., 2019, Paty and Cuturi, 2019, Lin et al., 2020, Nadjahi et al., 2020, Nietert et al., 2022, Genans and Wintenberger, 2026, Facco et al., 2017]. This paper places those geometries inside a common temporal-validity law. In that law, dimensionality matters only through the effective finite-sample exponent: raw multivariate Wasserstein can force slow rates, while projection, intrinsic-dimension, or smoothing mechanisms can preserve a nondegenerate horizon.

Age and drift. Wasserstein nonstationarity measures and Age-of-Information formulations make age into an operational variable [Jiang et al., 2020, Keehan et al., 2025, Amini and Vinas, 2026, Kaul et al., 2012, Yates et al., 2021]. The temporal-validity horizon gives that variable a distributional interpretation: age is costly because it accumulates geometric mismatch with the present.

Lag-geometry inference. Lag-geometry inference under noisy transport observations is complementary to the structural law. Parreira [2026b] study power-law lag geometry through a parameter-coupled heteroscedastic regression, deriving an information law for estimating (H, ζ) and testing the lag model itself.

7 Limitations and Further Directions

The closed theorem line covers the temporal-validity horizon, its useful-memory geometry, the one-dimensional proof anchor, the Gaussian lower and benchmark results, the embedded Sinkhorn theorems, and horizon-inference guarantees.

Closed theory. The fully proved finite-sample instantiation is the one-dimensional root-n regime under bounded support and fixed span. The structural lower bound is subclass based, and the exact Gaussian constants are internal to the available proof scheme rather than class-tight across broader deterministic Hölder path classes. Finite-family projection transfer is theorem-level under uniform projected control, but richer dominated families and raw multivariate Wasserstein lie beyond the proved range. In particular, no theorem here closes a general high-dimensional horizon law for raw multivariate Wasserstein, where the effective finite-sample exponent can become too small for practical memory selection.

Embedded Sinkhorn. For fixed-ε Sinkhorn, the compact-support one-dimensional pointwise and bandwise inheritance results are explicit, and bounded-cardinality finite discrete embedded classes satisfy the same moderate-band root-n law under a uniform diameter bound and a positive mass floor. The calibrated support-growth family is also closed on the discrete $k = 2$ grid. What remains open is support change beyond that fixed support-growth construction.

Empirical scope. The experiments target structural signatures: U-curves, roughness scaling, real-stream retention, and validity-detection separation. They do not identify detector-uniform delay laws or competitive end-to-end performance.

Further directions.

(1) **High-dimensional horizon geometry.** Identify which transport geometries in higher dimension preserve a nondegenerate temporal-validity horizon by keeping the effective finite-sample exponent away from zero. The main candidates are intrinsic-dimension, smoothed, and projection-robust formulations.

(2) **Sequential validity versus detection delay.** Quantify the structural gap between the validity horizon and the time-uniform detection horizon under false-alarm control, and determine how that gap scales across detector classes.

(3) **Adaptive horizon inference.** Develop stable estimators of the roughness and amplitude parameters that drive the horizon law, with explicit guarantees for memory localization under lag noise.

(4) **Support-changing embedded Sinkhorn.** Extend the calibrated embedded Sinkhorn theorem to classes with changing support and beyond the fixed support-growth construction.

8 Conclusion

Under drift, memory is never purely beneficial. It reduces finite-sample noise, but it also ages. The resulting tradeoff makes finite memory a temporal-validity object with a finite horizon.

The horizon law

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(n)}, P_t\Big) \le C_K n^{-a} + C_S \zeta n^H,$$

and its optimizer

$$n^*(a,H) \asymp (C_K/\zeta)^{1/(a+H)}$$

summarize that balance. This law induces a useful-memory region around the optimal scale.

The core theory closes in three theorem-level realizations of the same law: a tractable one-dimensional root-n proof anchor, bounded-cardinality finite- discrete embedded Sinkhorn, and a calibrated support-growth embedded Sinkhorn family. A structural Gaussian lower bound and a Gaussian location benchmark identify the horizon as a property of drift geometry rather than of a particular estimator.

The empirical signatures align with the theory: finite-memory U-curves, roughness-dependent horizon scaling, a real-stream finite-memory pattern, and a positive lag between validity loss and detector alarm. Horizon inference adds a second distinction: a temporal-validity horizon can exist structurally and still be operationally hard to estimate from lag geometry alone.

Further directions concern support-changing embedded Sinkhorn classes, high-dimensional horizon geometry, sequential validity versus detection delay, adaptive horizon inference, non-uniform memory design, and online control.

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Proof Details

This appendix gives the detailed derivations for structural results whose full calculations would interrupt the main line.

A.1 Gaussian lower-bound constants

Let

$$S_h := \sum_{r=1}^{h} r^{2H} \qquad \text{and} \qquad G_h := \sum_{r=1}^{h} \big(h^H - (h-r)^H\big)^2.$$

For the Gaussian two-point model with equal priors and common covariance $\sigma^2 I_h$, the Bayes classification error between means $\pm m$ is $\Phi(-\|m\|/\sigma)$. Therefore the ramp and endpoint-minimal payoffs are exactly

$$L_h^{\mathrm{ramp}}(\sigma,\zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma}\sqrt{S_h}\right)$$

and

$$L_h^{\min}(\sigma,\zeta) = \zeta h^H \Phi\left(-\frac{\zeta}{\sigma}\sqrt{G_h}\right).$$

Proposition A.1 (Gaussian lower-bound constants)**.** *Fix $H \in (0,1]$ and set $p_H := 2H/(2H+1)$. Let $x_H > 0$ be the unique maximizer of $x^{p_H}\Phi(-x)$, equivalently the unique solution of*

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

Then, as $\sigma/\zeta \to \infty$,

$$\sup_{h\ge 1} L_h^{\mathrm{ramp}}(\sigma,\zeta) \sim C_H^{\mathrm{ramp}}\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\mathrm{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Among all endpoint-saturating Hölder profiles, the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes the testing energy and has asymptotic energy constant

$$I_H = \int_0^1 \big(1-(1-u)^H\big)^2\, du = \frac{2H^2}{(H+1)(2H+1)}.$$

Moreover,

$$\sup_{h\ge 1} L_h^{\min}(\sigma,\zeta) \sim C_H^{\min}\, \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)},$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H),$$

and for $H < 1$ the improvement factor is

$$\frac{C_H^{\min}}{C_H^{\mathrm{ramp}}} = \left(\frac{H+1}{2H^2}\right)^{H/(2H+1)} > 1.$$

Proof of proposition A.1. Fix $H \in (0,1]$ and write $p_H = 2H/(2H+1)$.

For the ramp profile,

$$S_h \sim \frac{h^{2H+1}}{2H+1}.$$

Set

$$h = A\left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)}$$

with $A > 0$ fixed. Then

$$\frac{\zeta}{\sigma}\sqrt{S_h} \to \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

At the same time,

$$\zeta h^H = A^H \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}.$$

Hence

$$\frac{L_h^{\mathrm{ramp}}(\sigma,\zeta)}{\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}} \to A^H \Phi\left(-\frac{A^{H+1/2}}{\sqrt{2H+1}}\right).$$

Define

$$x = \frac{A^{H+1/2}}{\sqrt{2H+1}}.$$

Then

$$A^H = (2H+1)^{H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$(2H+1)^{H/(2H+1)} x^{p_H} \Phi(-x).$$

Its logarithmic derivative is

$$\frac{p_H}{x} - \frac{\varphi(x)}{\Phi(-x)}.$$

Since the inverse Mills ratio $\varphi(x)/\Phi(-x)$ is strictly increasing on $(0,\infty)$, this objective has a unique maximizer $x_H > 0$, characterized by

$$p_H \Phi(-x_H) = x_H \varphi(x_H).$$

This proves

$$\sup_{h\geq 1} L_h^{\mathrm{ramp}}(\sigma,\zeta) \sim C_H^{\mathrm{ramp}} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\mathrm{ramp}} = (2H+1)^{H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

For the endpoint-minimal profile, the Hölder constraint and endpoint condition imply

$$g_r \geq g_h - (h-r)^H = h^H - (h-r)^H$$

for every feasible endpoint-saturating profile g. Therefore the profile

$$g_r^{\min} = h^H - (h-r)^H$$

minimizes testing energy pointwise and hence minimizes $\sum_{r=1}^h g_r^2$. Its rescaled energy satisfies the Riemann-sum limit. Thus

$$h^{-(2H+1)} G_h \to I_H,$$

where

$$I_H = \int_0^1 \left(1-(1-u)^H\right)^2 du = \frac{2H^2}{(H+1)(2H+1)}.$$

With the same scaling

$$h = A\left(\frac{\sigma}{\zeta}\right)^{2/(2H+1)},$$

one has

$$\frac{\zeta}{\sigma}\sqrt{G_h} \to A^{H+1/2}\sqrt{I_H}.$$

Setting $x = A^{H+1/2}\sqrt{I_H}$ gives

$$A^H = I_H^{-H/(2H+1)} x^{p_H},$$

so the normalized limiting payoff becomes

$$I_H^{-H/(2H+1)} x^{p_H} \Phi(-x).$$

The same maximizer x_H therefore yields

$$\sup_{h\geq 1} L_h^{\min}(\sigma,\zeta) \sim C_H^{\min} \sigma^{2H/(2H+1)} \zeta^{1/(2H+1)}$$

with

$$C_H^{\min} = I_H^{-H/(2H+1)} x_H^{2H/(2H+1)} \Phi(-x_H).$$

Finally,

$$\frac{C_H^{\min}}{C_H^{\mathrm{ramp}}} = \left(\frac{I_H^{-1}}{2H+1}\right)^{H/(2H+1)} = \left(\frac{H+1}{2H^2}\right)^{H/(2H+1)}.$$

This factor is strictly larger than 1 for $H < 1$ and equal to 1 at $H = 1$.

A.2 Two-point lower bound used in proposition 2.10

The structural lower bound above is the exponent-level consequence of the exact Gaussian two-point calculation. For the ramp pair

$$\mu_{t-j}^{\pm,h} = \pm\zeta(h-j)_+^H,$$

the present-time separation is $2\zeta h^H$, so any estimator incurs risk at least of order $L_h^{\mathrm{ramp}}(\sigma,\zeta)$. Minimizing over h and using the asymptotic just proved gives the lower scale

$$\sigma^{2H/(2H+1)}\zeta^{1/(2H+1)}$$

and therefore the horizon exponent $(2H+1)^{-1}$ for the root-n regime.

A.3 Weighted-memory summaries

For a finite-memory estimator with weights $w_0, \ldots, w_{n-1}$, define the effective sample size

$$n_{\mathrm{eff}}(w) := \left(\sum_{j=0}^{n-1} w_j^2\right)^{-1}$$

and the weighted lag moment

$$M_H(w) := \sum_{j=0}^{n-1} j^H w_j.$$

Any weighted horizon law of the form

$$\mathbb{E}\, d\Big(\widehat{P}_t^{(w)}, P_t\Big) \le C_K n_{\mathrm{eff}}(w)^{-a} + C_S \zeta M_H(w)$$

balances the finite-sample and staleness terms through these two summaries.

Proposition A.2 (Uniform weights maximize effective sample size)**.** *Let $w_0, \dots, w_{n-1}$ be nonnegative weights with $\sum_{j=0}^{n-1} w_j = 1$. Then*

$$n_{\mathrm{eff}}(w) \le n,$$

with equality if and only if $w_j = 1/n$ for all j. Moreover, if the drift profile is an exact linear ramp in the sense that the lag-j contribution is proportional to j, then the weighted staleness equals the effective lag.

Proof. By Cauchy–Schwarz,

$$1 = \Big(\sum_{j=0}^{n-1} w_j\Big)^2 \le n \sum_{j=0}^{n-1} w_j^2,$$

so $n_{\mathrm{eff}}(w) = 1/\sum_j w_j^2 \le n$, with equality exactly when the weights are all equal. On an exact linear ramp, averaging the lag contribution with weights w_j recovers the weighted mean lag.

B Finite-Sample and Operational Diagnostics

This appendix gives finite-sample diagnostics for the theorem line and the operational implications.

B.1 Finite-Sample Geometry Diagnostic

For the finite-sample term, we estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in n. We also probe the fixed-span one-dimensional triangular window directly by comparing a triangular-array sample, with one draw from each drifted component, against an i.i.d. sample from the same window mixture. In that diagnostic the within-window span ζn^H is held fixed. Finally, we probe the Sinkhorn diagnostic at a few fixed values of ε.

In low dimension, the raw i.i.d. Wasserstein term is compatible with a root-n rate, whereas higher-dimensional raw Wasserstein slows down. In the one-dimensional fixed-span triangular-array diagnostic, the recovered slopes stay near the corresponding i.i.d. mixture baseline, which is consistent with the proof-model and design-effect reading used in the 1D proof-model proposition.

B.2 Bahadur Remainder Diagnostics

Two deformation families make the proof-model remainder assumption concrete. The first is a smooth fixed-support monotone warp class $g_\delta(x) = x + \delta x(1-x)$ with $|\delta| \le \Delta < 1$. On the tested spans 0.2 and 0.4, the integrated remainder decays at rates between 1.40 and 1.54, and the scaled quantity $n \int_0^1 R_n(u)^2 du$ still decays at rates between 0.40 and 0.54. The second is the fixed-span triangular-array uniform-kernel model used in the proof line. There the integrated residual rate is about 1.43 for the triangular sample and 1.49 for the i.i.d. mixture benchmark, while the residual-to-MSE ratio drops from about 0.38 at $n = 25$ to about 0.11 at $n = 400$.

These diagnostics locate the role of drift span explicitly. Within the tested fixed-span range, the remainder stays lower-order and the empirical rate remains well above the n^{-1} threshold needed for item (iv). Rougher kernels and growing-span experiments remain less regular, which is why the analysis keeps the general triangular-array verification problem open.

Table 2: Empirical scaling diagnostic for the finite-sample term. The bounded-support i.i.d. W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, the corresponding triangular-array diagnostic compares one sample from each drifted component against an i.i.d. sample from the window mixture, and the Sinkhorn diagnostic is reported at fixed ε. These rows are numerical support for the operational finite-sample discussion rather than theorem statements.

Setting	Estimated slope	Comment
W_2, $d = 1$	-0.48	close to $-1/2$
W_2, $d = 2$	-0.41	slower decay
W_2, $d = 4$	-0.25	slower decay
W_2, $d = 5$	-0.24	slower decay
W_2, i.i.d. mixture, $d = 1$, $H = 0.5$	-0.50	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 0.5$	-0.45	fixed span $\zeta n^H = 0.25$
W_2, i.i.d. mixture, $d = 1$, $H = 1.0$	-0.47	fixed span $\zeta n^H = 0.25$
W_2, triangular, $d = 1$, $H = 1.0$	-0.41	fixed span $\zeta n^H = 0.25$
Sinkhorn $\varepsilon = 0.50$	-0.21	stability constant changes
Sinkhorn $\varepsilon = 0.10$	-0.53	stability constant changes
Sinkhorn $\varepsilon = 0.05$	-0.67	stability constant changes

B.3 Extended and Operational Regimes

The extended rows compare triangular and i.i.d. mixture rates on embedded low-intrinsic-dimensional support across span. The operational rows compare fixed-ε Sinkhorn rates across regularization values on embedded low-intrinsic-dimensional support in ambient dimension eight.

Table 3: Representative rows for the extended and operational finite-sample regimes. The extended rows compare triangular and i.i.d. mixture rates across span, while the operational rows compare fixed-ε Sinkhorn across ε on embedded low-intrinsic-dimensional support.

Regime	Setting	Parameter	Rate exponent a
extended	triangular ambient d=8, intrinsic k=1, span=0.10	0.10	0.4652
extended	iid mixture ambient d=8, intrinsic k=1, span=0.10	0.10	0.5722
extended	triangular ambient d=8, intrinsic k=1, span=0.25	0.25	0.4737
extended	iid mixture ambient d=8, intrinsic k=1, span=0.25	0.25	0.5588
extended	triangular ambient d=8, intrinsic k=1, span=0.50	0.50	0.4792
extended	iid mixture ambient d=8, intrinsic k=1, span=0.50	0.50	0.5425
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5599
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.50	0.50	0.5706
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.20	0.20	0.5466
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.20	0.20	0.6150
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.10	0.10	0.4646
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.10	0.10	0.5060
operational	Sinkhorn iid mixture ambient d=8, intrinsic k=1, eps=0.05	0.05	0.4339
operational	Sinkhorn triangular ambient d=8, intrinsic k=1, eps=0.05	0.05	0.5124

The extended-regime rows indicate that low intrinsic dimension preserves a rate near the root-n regime without a dramatic triangular-versus-benchmark gap. The operational-regime rows record empirical evidence relevant to the fixed- ε Sinkhorn setting in definition 2.14: together with exact support-complexity inheritance and the dual threshold $\alpha > k/2$, they align with the embedded bandwise statement. The bounded- cardinality finite-support theorem closes one theorem-level subclass; the remaining question is support growth and support change.

Two complementary diagnostics sharpen the same boundary. First, direct probes of the centered Sinkhorn scaling Jacobian on the embedded fixed-span model show that bands approaching zero are the numerically unstable region, whereas on a moderate band such as $[0.2, 0.8]$ the cross-coupling block remains contractive and the self-coupling inverse norms appear to plateau with sample size. Second, on the supported synthetic grid, intrinsic-dimension estimates predict the observed stable-band cutoff better than ambient dimension. These diagnostics identify a useful regime statistic and isolate the band where the instability is concentrated. They probe a larger support-growth regime than the bounded-cardinality theorem.

The calibrated summary separates three ingredients on the same embedded grid: the empirically stable band, the certified band obtained from the structural certificate, and the self-coupling proxy on the calibrated pairs where that proxy is currently informative. On $(8, 2)$ and $(12, 2)$, the empirical and certified bands both reach 0.5, and the two self-coupling blocks keep positive spectral gap, worst spectral radius below 0.97, and largest-sample mean inverse norm below 3.0.

Table 4: Moderate-band Sinkhorn summary on the calibrated embedded grid. The table reports the empirical stable band, the certified band, the minimum recovered finite-sample rate exponents inside the certified band, and whether both self-coupling blocks satisfy the spectral-radius and inverse-norm proxy thresholds.

Pair	Emp. $\varepsilon_{\max}$	Thm.-ready $\varepsilon_{\max}$	Min iid a	Min tri. a	Self
$(8, 1)$	0.5	0.5	0.488	0.458	–
$(8, 2)$	0.5	0.5	0.455	0.419	yes
$(12, 1)$	0.2	0.2	0.501	0.474	–
$(12, 2)$	0.5	0.5	0.428	0.493	yes

Table 5: Calibrated moderate-band Sinkhorn certificates on the embedded fixed-span grid. The table reports the largest regularization value for which the certificate returns a certified band on the tested grid.

Ambient / intrinsic	α	Max certified ε
$(8, 1)$	2.0	0.5
$(8, 2)$	2.0	0.5
$(12, 1)$	2.0	0.2
$(12, 2)$	2.0	0.5

On the calibrated discrete $k = 2$ support grid, the support-growth diagnostics sharpen that picture further. The centered self-coupling operator for the squared block keeps radius well below one across the moderate band, the exact-target null Sinkhorn slopes are close to the n^{-1} remainder law, and the finite-support root-n empirical fluctuations remain bounded on the tested range. These diagnostics support the corrected S^2 self-coupling route and the finite- dimensional influence argument on the calibrated model. The analytic route is a finite-state minorization on the compact support rectangle: the kernel is uniformly bounded below on the calibrated band, so the centered block is a strict contraction and the inverse bound follows from a Doeblin-type decomposition. The support-growth probe points in the same direction: the ratio between exact-target Sinkhorn cost and squared ℓ_2 weight deviation stays uniformly small on the reported grid, with mean values below 0.084 and worst pointwise values below 0.24. Together with the exact $3/(4n)$ calibrated ℓ_2 fluctuation identity and the Hessian criterion in the main text, this organizes the support-growth family around a covariance-weighted trace bound and a local block-curvature bound for the null Sinkhorn map. A finite-difference basis probe supports that organization numerically: across the reported grid, local blockwise curvatures stay below about 0.24 and the stiffer collective directions scale approximately like $c(\varepsilon)n$ with reduced-channel coefficient between about 0.04 and 0.13. The local-basis trace proxy grows linearly in n at about 0.34 per block, while the covariance-weighted proxy stays around 0.002 to 0.004.

Table 6: Calibrated support-growth Sinkhorn diagnostics on the discrete $k = 2$ support grid with span 0.25 and moderate band $\mathcal{B} = [0.1, 0.5]$. The table reports the worst centered self-coupling radius for the squared operator, the corresponding inverse-norm bound, the ranges of the exact-target null slopes for triangular and i.i.d. samples, and the worst root-n support-deviation ratio across the band.

Pair	Worst $\rho(S_\varepsilon^2)$	Worst $\|(I - S_\varepsilon^2)^{-1}\|$	Tri slope range	IID slope range	Worst root-n ratio
$(8, 2)$	0.116	1.132	$[-1.001, -0.992]$	$[-0.880, -0.878]$	2.582
$(12, 2)$	0.116	1.132	$[-0.878, -0.826]$	$[-0.843, -0.778]$	2.840

B.4 Smooth Multivariate Transfer Diagnostics

The multivariate transfer diagnostic uses a drifting three-dimensional mean vector and a smooth log-sum-exp functional. The finite-sample error of the window estimator, the drift-induced bias of the window target, and the linearization residual are tracked separately across window lengths. The target behavior is the same as in the abstract multivariate transfer lemma: the finite-sample piece decreases with window length, the drift piece increases, and the linearization residual remains lower order than the finite-sample term on the tested windows.

Window	Tot.	Fin.	Drift	Resid.
8	0.2009	0.1988	0.0165	0.0239
16	0.1513	0.1542	0.0355	0.0162
32	0.1373	0.1363	0.0735	0.0066
64	0.1492	0.0807	0.1503	0.0051

Table 7: Smooth multivariate transfer diagnostic under triangular-array drift. The finite-sample term decreases with window length, the drift term increases, the total error has an interior minimum, and the linearization residual stays lower order across the tested windows.

B.5 Contribution Summary

(1) Object: temporal-validity horizon and useful-memory region. Finite-memory distribution tracking under drift is governed by a temporal-validity horizon together with a useful-memory region whose normalized geometry is determined by (a, H).

(2) Theorem line: proof anchor and closed classes. A one-dimensional root-n proof anchor, exact uniform-window staleness control, Gaussian lower/benchmark results, and theorem-level embedded Sinkhorn closures show that the horizon is structural across tractable and operational geometries.

(3) Transfer: projection families and Sinkhorn geometry. Finite projection and sliced geometries yield transfer theorems, and the bounded-cardinality finite-discrete and calibrated support-growth embedded Sinkhorn classes satisfy theorem-level horizon laws.

(4) Inference: identifiability and regret. Plug-in horizon inference yields an identifiability criterion for useful-memory selection and a quadratic regret law for horizon mislocation.

(5) Empirics: signatures of temporal validity. Empirical results exhibit interior U-curves, roughness-dependent scaling, real-stream finite-memory diagnostics, and pre-detection validity loss.

B.6 Robustness Summaries

The real-stream, validity-gap, and online-controller sweeps are summarized here in compact form.

Variable	Runs	Interior	Best-window range	Edge/best range
`nswprice`	9	9/9	14–435	1.31–1.68
`nswdemand`	9	9/9	25–327	1.15–3.24

Table 8: ELEC2 robustness summary across anchor sizes 24, 48, and 96 and segment starts 0, 8000, and 16000, with segment length 4000. The interior-optimum count records how often the best window stays away from the search boundary; the edge/best range reports the ratio between the worse boundary error and the optimum.

Detector input	Detector	Detection rate	Mean-gap range	Excess-area range
Observation	ADWIN	1.00	429–620	7.41–20.41
Observation	Page–Hinkley	1.00	205–345	3.55–8.20
Abs. residual	ADWIN	0.00	—	—
Abs. residual	Page–Hinkley	0.33–0.92	994–1665	16.94–151.24

Table 9: Validity-detection robustness summary across operating windows 160, 220, and 300 and the reported threshold grid. Mean-gap and excess-area ranges are computed over the successful runs in each detector/input family.

Profile	Best static/oracle	Activity/oracle	Plug-in/oracle	Structural/oracle	Adaptive/oracle
Default	1.43	1.36	4.41	1.36	1.30
Smooth	1.05	1.41	3.24	1.22	1.24
Rough	1.09	1.65	2.59	1.30	1.39
Alternating	1.02	1.65	2.30	1.24	1.36
Ramp-up	1.02	1.34	2.91	1.18	1.23

Table 10: Online memory-selection baselines across five profile families, aggregated over seeds $0, 1, 2$, with maximum one-step window-index jumps. Entries report mean absolute error relative to the oracle controller. The structural controller consistently improves on the raw plug-in and activity baselines and stays close to oracle performance across profile changes.

Scheme	Best window	MAE	n_{eff}	Mean lag	Horizon proxy
Uniform	12	0.272	12.00	5.50	3.880
Triangular	18	0.257	13.86	5.67	3.941
Geometric	15	0.258	12.15	4.95	3.607

Table 11: Weight-sensitivity summary for uniform, triangular, and geometric tapers. The table reports the best window under mean absolute error together with the induced effective sample size, mean lag, and the horizon proxy $n_{\mathrm{eff}}^{-a} + M_H$ used in the diagnostic. Geometric tapering reduces lag most strongly, while triangular tapering enlarges effective sample size at a higher lag cost.

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